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Converting telemetry values into common data formats in a processor-based system in an integrated circuit (IC) chip

Abstract

Converting telemetry values into common data formats in a processor-based system in an integrated circuit (IC) chip is disclosed herein. In one aspect, an IC chip comprises a processor-based system that is configured to receive an input telemetry value from an input source circuit. The processor-based system converts the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10. The processor-based system then processes common format telemetry value.

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Background/Summary

BACKGROUND

I. Field of the Disclosure

(1) The field of the disclosure relates to processor-based systems (e.g., a central processing unit (CPU)-based system, a graphic processing unit (GPU)-based system, or a neural network processing unit (NPU)-based system), and more particularly to a common data format in a processor-based system.

II. Background

(2) Microprocessors, also known as processing units (PUs), perform computational tasks in a wide variety of applications. One type of conventional microprocessor or PU is a central processing unit (CPU). Another type of microprocessor or PU is a dedicated processing unit known as a graphics processing unit (GPU). A GPU is designed with specialized hardware to accelerate the rendering of graphics and video data for display. A GPU may be implemented as an integrated element of a general-purpose CPU, or as a discrete hardware element that is separate from the CPU. Other examples of PUs may include neural network processing units or neural processing units (NPU). In cases of PUs, the PUs are configured to execute software instructions that instruct a processor to fetch data from a location in memory, and to perform one or more processor operations using the fetched data.

(3) PUs are included in a computer system that includes other supporting processing devices (circuits) involved with or accessed as part of performing computing operations in the computer system. Examples of these other supporting processing devices include memory, input/output (I/O) devices, secondary storage, modems, video processors, and related interface circuits. The PUs and supporting processing devices are referred to collectively as processing devices. Processing devices of a processor-based system can be provided in separate ICs in separate IC chips. Alternatively, processing devices of a processor-based system can also be aggregated in a larger IC, like a system-on-a-chip (SoC) IC, wherein some or all of these processing devices are integrated into the same IC chip. For example, a SoC IC chip may include a PU that includes a plurality of processor cores and supporting processing devices such as a memory system that includes cache memory and memory controllers for controlling access to external memory, I/O interfaces, power management systems, etc. A SoC may be particularly advantageous for applications in which a limited area is available for the computer system (e.g., a mobile computing device such as cellular device). To manage power distributed to the processing devices, the SoC may also include a power management system that includes one or more power rails in the SoC that supply power to its components. A separate power management integrated circuit (PMIC) that can be on- or off-chip with the SoC can independently control power supplied to the power rails. The SoC may be designed with a plurality of different power rails that are distributed within the SoC to provide power to various clusters of the processing devices for their operation. For example, all the processor cores in the SoC may be coupled to a common power rail for power, whereas supporting processing devices may be powered from separate power rails in the SoC, depending on the design of the SoC.

SUMMARY OF THE DISCLOSURE

(4) Aspects disclosed herein include converting telemetry values into common data format in a processor-based system. Further aspects include hierarchical power estimation and throttling in a processor-based system in an integrated circuit (IC) chip that makes use of the common data format. Related power management and power throttling methods are also disclosed. The IC chip includes a processor as well as integrated supporting processing devices (e.g., network nodes, memory controllers, internal memory, input/output (I/O) interface circuits, etc.) for the processor. For example, the processor may be a central processing unit (CPU), graphics processing unit (GPU) or a neural network processing unit (NPU), wherein the processor includes multiple processing units (PUs) and/or processor cores. The processor-based system may be provided as a system-on-a-chip (SoC) that includes a processor and the integrated supporting processing devices for the PU. As examples, the SoC may be employed in smaller, mobile devices (e.g., a cellular phone, a laptop computer), as well as enterprise systems such as server chips in computer servers. The IC chip also includes a hierarchical power management system that is configured to control

power consumption by the processor-based system at both local and centralized levels to achieve a desired performance within an overall power budget for the IC chip. The hierarchical power management system can be configured to control power consumption by controlling the power level (e.g., voltage level) distributed at one or more power rails in the IC chip that provide power to the PUs and the integrated supporting processing devices. For example, the hierarchical power management system can be configured to provide additional power to certain power rails supplying power to higher current demanding devices to achieve higher performance, while providing less power to other power rails to keep the overall power within power and/or thermal limits for the IC chip. The hierarchical power management system can also be configured to control power consumption by throttling performance (e.g., frequency) of the processing devices in the processor-based system, which in turn throttles (i.e., reduces, maintains, or increases) their current demand and thus their power consumption. Note as used herein, throttle can mean to take an action that will decrease or increase a parameter that affects power and thus results in a respective decrease or increase in power consumption.

(5) The hierarchical power management system is configured to throttle performance of the processing devices in the processor-based system, because the level of processing activity in the processing devices in a SoC can vary based on workload conditions. Some power rails in the SoC may experience heightened current demand. It is desired that this current demand not exceed the maximum current limitations of its respective power rail. Even if a higher current demand on a power rail is within its maximum current limits, a heightened activity of a processing device in the SoC can generate a sudden increase in current demand from its power rail, referred to as a “di/dt” event. This di/dt event can cause a voltage droop in the power rail, thus negatively affecting performance of processing devices powered by such power rail. Also, even if a higher current demand on a power rail is within its maximum current limits, a higher current demand can increase the overall power consumption of the SoC. Processing devices may have a maximum power rating to properly operate and/or to not impact performance in an undesired manner. Higher current demand from processing devices can also generate excess heat. Thus, the maximum power rating of the SoC may be based in part on the ability of the SoC to dissipate heat generated by the processing devices during their operation.

(6) In exemplary aspects, the hierarchical power management system includes local area management (LAM) circuits distributed in the IC chip that are each associated with one or more processing devices in the IC chip. The LAM circuits are configured to generate power events associated with its monitored processing devices in the IC chip that represent power consumption associated with the monitored processing devices in the IC chip. The power events can be reported from local areas in the IC chip where power estimations for particular monitored processing devices are performed, to a centralized power estimation and limit (PEL) circuit in the hierarchical power management system. The PEL circuit is configured to estimate and control (i.e., throttle) power in the processor-based system in the IC chip to achieve a desired performance within an overall power budget for the IC chip. The PEL circuit determines how to throttle power based on the received power events. For example, the power events may be associated with estimations of power consumption that can be thought of as power throttle recommendations to throttle power in the IC chip if the estimated power consumption exceeds the power limits of the IC chip or negatively affects performance.

(7) The activity of the processing devices in the IC chip affects its steady state and transient current (i) demand (di/dt), and thus its power consumption. Because the IC chip may be larger in terms of die area due to the integration of the PUs and integrated supporting processing devices, there can be significant delay between when the PEL circuit receives a power event regarding power consumption of a monitored processing device and the PEL circuit throttling power in the IC chip to throttle power consumption in response. This delay can, for example, cause devices in the IC chip to temporarily continue to consume excess power that can cause thermal and/or power issues

(e.g., di/dt issues, voltage droop, heat generation) before the power management circuit has time to react.

(8) Mechanisms for addressing the above-noted issues may face additional challenges because the various processing devices reporting telemetry data (e.g., power telemetry data reported to the PEL circuit) may employ different units, precisions, and dynamic ranges when measuring energy and/or electric charge. Accordingly, telemetry data received from different processing devices must be scaled and/or converted into common ranges and units, before processing the data (e.g., by aggregating the data from the different processing devices for estimation for longer time windows). However, such scaling and conversion operations may require the use of multiplier circuits and/or divider circuits, which may increase the complexity of the SoC, consume additional processor resources, and negatively affect the overall performance of the SoC.

(9) To convert telemetry values (e.g., data received as part of the received power events) into common data formats, the IC chip (e.g., using the PEL circuit in exemplary aspects disclosed herein) is configured to receive an input telemetry value from an input source circuit of the processor-based system of the IC chip. The input source circuit may comprise, as non-limiting examples, a LAM circuit of the IC, a regional activity management (RAM) circuit of the IC, a processing unit of the IC, or a power management IC (PMIC) of the IC. The IC chip converts the input telemetry value into the common format telemetry value. However, instead of performing multiplying or dividing the input telemetry value by a power of 10 to convert it into the common format telemetry value (which would require the use of a multiplier circuit and/or a divider circuit), the IC chip is configured to perform a bit shift operation on the input telemetry value, effectively multiplying or dividing the input telemetry value by a power of two (2). The common format telemetry value is then stored as a value wherein a unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of the power of two (2) and a corresponding power of 10. In some aspects, the input telemetry value comprises an input power telemetry value and the common format telemetry value is a common format power telemetry value, with the input power telemetry value and the common format power telemetry value each representing a power measurement (e.g., a measurement of one of capacitance, electric charge, energy, and current, as non-limiting examples).

(10) For instance, if the input telemetry value comprises a micro-unit to be converted into a milli-unit (i.e., by dividing the input telemetry value by 10^3 or 1,000), the IC chip is configured to convert the input telemetry value into the common format telemetry value by performing a rightward bit shift of the input telemetry value by 10 bits (i.e., dividing the input telemetry value by 2^{10} or 1,024), with the unit value of the resulting common format telemetry value being 1.024 based on the quotient of $2^{10}/10^3$. In aspects in which conversion from, e.g., a nano-unit to a milli-unit is required (i.e., by dividing the input telemetry value by 10^6 or 1,000,000), the IC chip is configured to convert the input telemetry value into the common format telemetry value by performing a rightward bit shift of the input telemetry value by 20 bits (i.e., dividing the input telemetry value by 2^{20} or 1,048,576), with the unit value of the resulting common format telemetry value being 1.048 based on the quotient of $2^{20}/10^6$. Aspects in which the input telemetry value comprises, for example, a pico-unit to be converted into a milli-unit (i.e., by dividing the telemetry value by 10^9 or 1,000,000,000), the IC chip is configured to convert the input telemetry value into the common format telemetry value by performing a rightward bit shift of the input telemetry value by 30 bits (i.e., dividing the input telemetry value by 2^{30} or 1,073,741,824), with the unit value of the resulting common format telemetry value being 1.073 based on the quotient of $2^{30}/10^9$.

(11) The IC chip may process the common format telemetry value. In some examples, the PEL circuit of the IC chip may aggregate the common format power telemetry value into an aggregated power telemetry value, by, e.g., summing the common format power telemetry value with a current value of the aggregated power telemetry value having the same unit value. The PEL circuit may

subsequently generate a power limiting management response to cause power consumption to be throttled in the IC chip based on the aggregated power telemetry value. In aspects in which an input source circuit may provide an input power telemetry value having a unit value that already matches the unit value of the aggregated power telemetry value (and thus may not require conversion), the PEL circuit may be configured to first determine whether a conversion of the input power telemetry value into a common format power telemetry value is required. In such aspects, the PEL circuit converts the input power telemetry value into the common format power telemetry value responsive to determining that the conversion of the input power telemetry value into the common format power telemetry value is required.

(12) While in exemplary examples of the present disclosure, the IC chip comprising a processor-based system is described as comprising a centralized PEL circuit, aspects of the present disclosure are not limited to IC chips and processor-based systems that comprise PEL circuits. Furthermore, while in exemplary examples of the present disclosure, the input telemetry value is an input power telemetry value, and the common format telemetry value is a common format power telemetry value, aspects of the present disclosure are not limited to telemetry values representing in power. Rather, the methods, apparatuses, computer programs and non-transitory computer-readable medium according to the present disclosure may be useful in any processor-based systems where values from different sources, having different units, and/or comprising unitless quantities may be stored and converted to a common format.

(13) In some aspects, a firmware of the processor-based system may receive an aggregated power telemetry value (e.g., as a result of the PEL circuit reporting the aggregated power telemetry value to the firmware). The firmware may perform post-processing to convert the aggregated power telemetry value to a converted power telemetry value, wherein a unit value represented by a least significant bit of the converted power telemetry value is one (1). This may be accomplished in some aspects by multiplying the aggregated power telemetry value by the unit value of the aggregated power telemetry value. For example, if the unit value of a least significant bit of the aggregated power telemetry value is 1.024, the firmware may convert the aggregated power telemetry to the converted power telemetry value by multiplying the aggregated power telemetry value by 1.024. The converted power telemetry value may then be used by the firmware for other post-processing operations and/or estimations, configuration of configuration state registers (CSRs), and the like.

(14) In another exemplary aspect, an IC chip is disclosed. The IC chip comprises a processor-based system. The processor-based system is configured to receive, from an input source circuit of the processor-based system, an input telemetry value. The processor-based system is further configured to convert the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10. The processor-based system is also configured to process the common format telemetry value.

(15) In another exemplary aspect, an IC chip is disclosed. The IC chip comprises means for receiving, from an input source circuit, an input telemetry value. The IC chip may further comprise means for converting the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10. The IC chip also comprise means for processing the common format telemetry value.

(16) In another exemplary aspect, a method for converting power telemetry values into common data formats in a processor-based system in an IC chip is disclosed. The method comprises receiving, from an input source circuit of the processor-based system, an input telemetry value. The method further comprises converting the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a

corresponding power of 10. The method also comprises processing the common format telemetry value.

(17) In another exemplary aspect, a non-transitory computer-readable medium is disclosed. The non-transitory computer-readable medium stores thereon computer-executable instructions that, when executed, cause a processor of a processor-based device to receive, from an input source circuit of the processor-based system, an input telemetry value. The computer-executable instructions further cause the processor to convert the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10. The computer-executable instructions also cause the processor to process the common format telemetry value.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 is a schematic diagram of an exemplary processor-based system in the form of an exemplary system-on-a-chip (SoC) in an integrated circuit (IC) chip;

(2) FIG. 2 is a logic diagram of the exemplary processor-based system in FIG. 1 illustrating processing circuits and other support devices communicatively coupled to an internal communication network, and a hierarchical power management system that is configured to perform power estimation and throttling of power consumption;

(3) FIG. 3 is a top view of an exemplary physical layout of the semiconductor die (“die”) that is an SoC in an IC chip of the processor-based system in FIG. 1 illustrating the physical layout of different tile regions and the devices physically present in such different tile regions, and coded with indicia indicating which devices are powered from separate power rails supplied with power from the hierarchical power management system;

(4) FIG. 4 is a table illustrating an exemplary assignment of power rails driven by respective power management ICs (PMIC) in the processor-based system in FIG. 1, to devices in the processor-based system for supplying power to such devices;

(5) FIG. 5 is another top view of the processor-based system in the IC chip in FIG. 1 illustrating local area management (LAM) circuits, regional area management (RAM) circuits, and a power estimation and limiting (PEL) circuit as part of a hierarchical power management system, wherein the hierarchical power management system is configured to locally monitor activity of devices in the processor-based system to estimate and throttle its power consumption and report activity power events regarding estimated power consumption to the PEL circuit, wherein the PEL circuit is configured to collect activity power events regarding power consumption of the monitored processing devices and throttle power in the IC chip in response;

(6) FIG. 6 is a schematic diagram of an exemplary three (3) level hierarchical power management system that can be provided in the processor-based system in the IC chip in FIG. 1, wherein the three (3) level hierarchical power management system includes: a first local level of LAM circuits configured to perform local device monitoring and power consumption throttling, and reporting activity power events regarding monitored processing device power consumption, a second, intermediate level of RAM circuits configured to receive and aggregate local activity power events, and a third, centralized level of a PEL circuit configured to collect aggregated activity power events regarding power consumption of the monitored processing devices and throttle power in the IC chip in response;

(7) FIG. 7 is a schematic diagram of another exemplary two (2) level hierarchical power management system that can be provided in the processor-based system in the IC chip in FIG. 1, wherein two (2) level hierarchical power management system includes: a first local level of LAM

circuits configured to perform local device monitoring and power consumption throttling, and a second, centralized level of a PEL circuit configured to collect aggregated activity power events regarding power consumption of the monitored processing devices and throttle power in the IC chip in response;

(8) FIG. 8 is a flowchart illustrating an exemplary process of the hierarchical power management system of the processor-based system in FIG. 6 locally monitoring and throttling power consumption of monitored processing devices and hierarchically reporting activity power events related to the monitored power consumption to a PEL circuit configured to throttle power consumption in the processor-based system in response to the received power events;

(9) FIG. 9A is a schematic diagram illustrating an exemplary di/dt circuit that can be provided in a LAM circuit in the hierarchical power management system in FIG. 6, wherein the di/dt circuit is configured to collect activity samples of a device monitored by the LAM circuit and correlate the activity samples to estimate current and generate estimated current samples in a designated time window which can then be used to determine the slope of a rate of change in current/current flow rate (di/dt) consumed by the monitored processing device to determine whether the power consumption of the monitored processing device should be throttled by the LAM circuit;

(10) FIG. 9B is a graph illustrating exemplary estimated current samples collected by the di/dt circuit in FIG. 9A plotted as a function of time to determine the rate of change in current flow rate (di/dt) consumed by the monitored processing device;

(11) FIG. 10 is a logic diagram of an exemplary PEL circuit that can be provided in the hierarchical power management system of the processor-based system, and that illustrates exemplary components for receiving power events, decoding the received power events in tracking circuits, and merging the tracked power events to generate a power limiting management responses to throttle power consumption in the processor-based system in response to the received power events;

(12) FIG. 11 is a block diagram illustrating an exemplary PEL circuit that is configured to convert input power telemetry values into common data format values where a least significant bit represents a unit value greater than one (1), according to some aspects;

(13) FIGS. 12A-12B provide a flowchart illustrating exemplary operations for converting input power telemetry values into common data format values where a least significant bit represents a unit value greater than one (1), according to some aspects;

(14) FIG. 13 is a block diagram of another exemplary processor-based system that includes a hierarchical power management system configured to locally monitor activity of devices in the processor-based system to estimate and throttle its power consumption and report activity power events regarding estimated power consumption to the PEL circuit, wherein the PEL circuit is configured to collect activity power events regarding power consumption of the monitored processing devices and throttle power in the IC chip in response, including but not limited to the hierarchical power management systems and their exemplary components in FIGS. 1-3, 5-7, 9A, 10, and 11; and

(15) FIG. 14 is a block diagram of an exemplary wireless communications device that includes a hierarchical power management system configured to locally monitor activity of devices in the processor-based system to estimate and throttle its power consumption and report activity power events regarding estimated power consumption to the PEL circuit, wherein the PEL circuit is configured to collect activity power events regarding power consumption of the monitored processing devices and throttle power in the IC chip in response, including but not limited to the hierarchical power management systems and their exemplary components in FIGS. 1-3, 5-7, 9A, 10, and 11.

DETAILED DESCRIPTION

(16) With reference now to the drawing figures, several exemplary aspects of the present disclosure are described. The word “exemplary” is used herein to mean “serving as an example, instance, or

illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects.

(17) Aspects disclosed herein include a hierarchical power estimation and throttling in a processor-based system in an integrated circuit (IC) chip. Related power management and power throttling methods are also disclosed. The IC chip includes a processor as well as integrated supporting processing devices (e.g., network nodes, memory controllers, internal memory, input/output (I/O) interface circuits, etc.) for the processor. For example, the processor may be a central processing unit (CPU), graphics processing unit (GPU) or a neural network processing unit (NPU), wherein the processor includes multiple processing units (PUs) and/or processor cores. The processor-based system may be provided as a system-on-a-chip (SoC) that includes a processor and the integrated supporting processing devices for the PU. As examples, the SoC may be employed in smaller, mobile devices (e.g., a cellular phone, a laptop computer), as well as enterprise systems such as server chips in computer servers. The IC chip also includes a hierarchical power management system that is configured to control power consumption by the processor-based system at both local and centralized levels to achieve a desired performance within an overall power budget for the IC chip. The hierarchical power management system can be configured to control power consumption by controlling the power level (e.g., voltage level) distributed at one or more power rails in the IC chip that provide power to the PUs and the integrated supporting processing devices. For example, the hierarchical power management system can be configured to provide additional power to certain power rails supplying power to higher current demanding devices to achieve higher performance, while providing less power to other power rails to keep the overall power within power and/or thermal limits for the IC chip. The hierarchical power management system can also be configured to control power consumption by throttling performance (e.g., frequency) of the processing devices in the processor-based system, which in turn throttles (i.e., reduces, maintains, or increases) their current demand and thus their power consumption. Note as used herein, throttle can mean to take an action that will decrease or increase a parameter that affects power and thus results in a respective decrease or increase in power consumption.

(18) The hierarchical power management system is configured to throttle performance of the processing devices in the processor-based system, because the level of processing activity in the processing devices in a SoC can vary based on workload conditions. Some power rails in the SoC may experience heightened current demand. It is desired that this current demand not exceed the maximum current limitations of its respective power rail. Even if a higher current demand on a power rail is within its maximum current limits, a heightened activity of a processing device in the SoC can generate a sudden increase in current demand from its power rail, referred to as a “di/dt” event. This di/dt event can cause a voltage droop in the power rail, thus negatively affecting performance of processing devices powered by such power rail. Also, even if a higher current demand on a power rail is within its maximum current limits, a higher current demand can increase the overall power consumption of the SoC. Processing devices may have a maximum power rating to properly operate and/or to not impact performance in an undesired manner. Higher current demand from processing devices can also generate excess heat. Thus, the maximum power rating of the SoC may be based in part on the ability of the SoC to dissipate heat generated by the processing devices during their operation.

(19) In exemplary aspects, the hierarchical power management system includes local area management (LAM) circuits distributed in the IC chip that are each associated with one or more processing devices in the IC chip. The LAM circuits are configured to generate power events associated with its monitored processing devices in the IC chip that represent power consumption associated with the monitored processing devices in the IC chip. The power events can be reported from local areas in the IC chip where power estimations for particular monitored processing devices are performed, to a centralized power estimation and limit (PEL) circuit in the hierarchical power management system. The PEL circuit is configured to estimate and control (i.e., throttle)

power in the processor-based system in the IC chip to achieve a desired performance within an overall power budget for the IC chip. The PEL circuit determines how to throttle power based on the received power events. For example, the power events may be associated with estimations of power consumption that can be thought of as power throttle recommendations to throttle power in the IC chip if the estimated power consumption exceeds the power limits of the IC chip or negatively affects performance.

(20) The activity of the processing devices in the IC chip affects its steady state and transient current (i) demand (i.e., change in current demand or current flow rate is often referred to as “di/dt”), and thus its power consumption. Because the IC chip may be larger in terms of die area due to the integration of the processing units and integrated supporting processing devices, there can be significant delay between when PEL circuit receives a power event regarding consumption of a monitored processing device and the PEL circuit throttling power in the IC chip to throttle power consumption in response. This delay can, for example, cause devices in the IC chip to temporarily continue to consume excess power that can cause thermal and/or power issues (e.g., di/dt issues, voltage droop, heat generation) before the power management circuit has time to react.

(21) Mechanisms for addressing the above-noted issues may face additional challenges because the various processing devices reporting power telemetry data to the PEL circuit may employ different units, precisions, and dynamic ranges when measuring energy and/or electric charge. Accordingly, the PEL circuit must scale and/or convert power telemetry data received from different processing devices and aggregated over different time windows into a common ranges and units, before aggregating the data from the different processing devices for estimation for longer time windows. However, such scaling and conversion operations may require the use of multiplier circuits and/or divider circuits, which may increase the complexity of the SoC, consume additional processor resources, and negatively affect the overall performance of the SoC.

(22) To convert power telemetry values (e.g., data received as part of the received power events) into common data formats, the PEL circuit in exemplary aspects disclosed herein is configured to receive an input power telemetry value from an input source circuit of the processor-based system of the IC chip. The input source circuit may comprise, as non-limiting examples, a LAM circuit of the IC, a regional activity management (RAM) circuit of the IC, a processing unit of the IC, or a power management IC (PMIC) of the IC. The input power telemetry value in some aspects may represent a measurement of one of capacitance, electric charge, energy, and current, as non-limiting examples. The PEL circuit converts the input power telemetry value into the common format power telemetry value. However, instead of performing multiplying or dividing the input power telemetry value by a power of 10 to convert it into the common format power telemetry value (which would require the use of a multiplier circuit and/or a divider circuit), the PEL circuit is configured to perform a bit shift operation on the input power telemetry value, effectively multiplying or dividing the input power telemetry value by a power of two (2). The common format power telemetry value is then stored as a value wherein a unit value represented by a least significant bit of the common format power telemetry value is greater than one (1) and is based on a quotient of the power of two (2) and a corresponding power of 10.

(23) For instance, if the input power telemetry value comprises a micro-unit to be converted into a milli-unit (i.e., by dividing the input power telemetry value by 10^3 or 1,000), the PEL circuit is configured to convert the input power telemetry value into the common format power telemetry value by performing a rightward bit shift of the input power telemetry value by 10 bits (i.e., dividing the input power telemetry value by 2^{10} or 1,024), with the unit value of the resulting common format power telemetry value being 1.024 based on the quotient of $2^{10}/10^3$. In aspects in which conversion from, e.g., a nano-unit to a milli-unit is required (i.e., by dividing the input power telemetry value by 10^6 or 1,000,000), the PEL circuit is configured to convert the input power telemetry value into the common format power telemetry value by performing a rightward bit shift of the input power telemetry value by 20 bits (i.e., dividing the input power

telemetry value by $2^{\text{sup.}20}$ or 1,048,576), with the unit value of the resulting common format power telemetry value being 1.048 based on the quotient of $2^{\text{sup.}20}/10^{\text{sup.}6}$. Aspects in which the input power telemetry value comprises, for example, a pico-unit to be converted into a milli-unit (i.e., by dividing the telemetry value by $10^{\text{sup.}9}$ or 1,000,000,000), the PEL circuit is configured to convert the input power telemetry value into the common format power telemetry value by performing a rightward bit shift of the input power telemetry value by 30 bits (i.e., dividing the input power telemetry value by $2^{\text{sup.}30}$ or 1,073,741,824), with the unit value of the resulting common format power telemetry value being 1.073 based on the quotient of $2^{\text{sup.}30}/10^{\text{sup.}9}$.

(24) The PEL circuit then aggregates the common format power telemetry value into an aggregated power telemetry value, by, e.g., summing the common format power telemetry value with a current value of the aggregated power telemetry value having the same unit value. The PEL circuit subsequently generates a power limiting management response to cause power consumption to be throttled in the IC chip based on the aggregated power telemetry value. In aspects in which an input source circuit may provide an input power telemetry value having a unit value that already matches the unit value of the aggregated power telemetry value (and thus may not require conversion), the PEL circuit may be configured to first determine whether a conversion of the input power telemetry value into a common format power telemetry value is required. In such aspects, the PEL circuit converts the input power telemetry value into the common format power telemetry value responsive to determining that the conversion of the input power telemetry value into the common format power telemetry value is required.

(25) In some aspects, a firmware of the processor-based system may receive the aggregated power telemetry value (e.g., as a result of the PEL circuit reporting the aggregated power telemetry value to the firmware). The firmware may perform post-processing to convert the aggregated power telemetry value to a converted power telemetry value, wherein a unit value represented by a least significant bit of the converted power telemetry value is one (1). This may be accomplished in some aspects by multiplying the aggregated power telemetry value by the unit value of the aggregated power telemetry value. For example, if the unit value of a least significant bit of the aggregated power telemetry value is 1.024, the firmware may convert the aggregated power telemetry to the converted power telemetry value by multiplying the aggregated power telemetry value by 1.024. The converted power telemetry value may then be used by the firmware for other post-processing operations and/or estimations, configuration of configuration state registers (CSRs), and the like.

(26) In this regard, FIG. 1 is a schematic diagram of an exemplary processor-based system **100** in the form of an exemplary system-on-a-chip (SoC) **102** in an integrated circuit (IC) chip **104** in which a hierarchical power management system can be provided. The SoC **102** may be employed in smaller, mobile devices (e.g., a cellular phone, a laptop computer), as well as enterprise systems such as server chips in computer servers. The processor-based system **100** is first described with regard to FIG. 1 before exemplary hierarchical power management systems that can be provided in the processor-based system **100** to estimate and throttle power consumption in the IC chip **104** are described starting at FIG. 2 below.

(27) With reference to FIG. 1, the processor-based system **100** is provided in a single semiconductor die **106** so that the processor-based system **100** is integrated into a single IC chip **104**. The processor-based system **100** includes a plurality of processing unit (PU) clusters **108(0)-108(N)** that are examples of processing devices **110** in the processor-based system **100**. The PU clusters **108(0)-108(N)** each can include one or more processor cores **112(0)-112(N)** each configured to execute instructions (e.g., software, firmware) to carry out tasks as is known for processors. For example, the PU clusters **108(0)-108(N)** may be central processing unit (CPU) clusters wherein one or more of the processor cores **112(0)-112(N)** includes CPUs and/or graphics processing unit (GPU) clusters, wherein one or more of the processor cores **112(0)-112(N)** includes GPUs. The processor-based system **100** includes an internal communication network **114** that

facilitates providing communication paths between the PU clusters **108(0)-108(N)** and other supporting processing devices, that are also considered processing devices, to carry out desired processing requests and related processing tasks. The PU clusters **108(0)-108(N)** are communicatively coupled to the internal communication network **114**. The internal communication network **114** can be a coherent communication bus that provides a fabric in the processor-based system **100**. The internal communication network **114** can be a network fabric that typically consists of network nodes and their communication lines, network of wires, and/or communication channels that provide communication paths that provide reliable communication between different PU clusters **108(0)-108(N)** and the supporting processing devices **110**. Network nodes are the circuits, such as interconnected switches and routers, that provide a reliable network fabric that provides and receives data on the communication paths between different PU clusters **108(0)-108(N)** and the supporting processing devices **110**. The fabric provided by the internal communication network **114** also includes a network of wires or communication channels that allow different processing devices in the processor-based system **100** to communicate and exchange data with each other at high speeds.

(28) For example, as shown in FIG. **1**, the processor-based system **100** also includes internal cache memory **116** and memory controllers (MCs) **118(0)-118(M)** as other types of processing devices **110** that provide access to memory. The cache memory **116** shown in FIG. **1** is a shared cache memory that is communicatively coupled to the internal communication network **114** and can be accessed by the PU clusters **108(0)-108(N)** through the internal communication network **114**. The processor-based system **100** may also include private cache memory and/or private shared cache memory that is integrated or privately accessible by one or more of the respective PU clusters **108(0)-108(N)** without having to access such through the internal communications network **114**. The memory controllers **118(0)-118(M)** are communicatively coupled to the internal communication network **114** in the IC chip **104**. The memory controllers **118(0)-118(M)** provide the PU clusters **108(0)-108(N)** access to memory for storing and retrieving data to carry out processing tasks. For example, the memory controllers **118(0)-118(M)** may be coupled to external memory from the IC chip **104** or internal memory integrated in the IC chip **104**.

(29) Also as shown in FIG. **1**, the processor-based system **100** in this example also includes I/O interface circuits **120(0)-120(X)** as other examples of processing devices **110** that are also communicatively coupled to the internal communication network **114**. The I/O interface circuits **120(0)-120(X)** provide access to I/O devices, which may be internal and integrated in the IC chip **104** or external to the IC chip **104**. For example, the I/O interface circuits **120(0)-120(X)** may be peripheral component interconnect (PCI) interface circuits that are used for connecting I/O hardware devices to a processor-based system, like the processor-based system **100** in FIG. **1**, to allow high-speed data to be transferred between devices and the PU clusters **108(0)-108(N)** in the processor-based system **100**.

(30) Also as shown in FIG. **1**, the processor-based system **100** in this example also includes socket-to-socket (S2S) interface circuits **122(0)-122(Y)** as other examples of processing devices **110** that are also communicatively coupled to the internal communication network **114**. The S2S interface circuits **122(0)-122(Y)** allow the processor-based system **100** to be coupled to another separate processor-based system (which may be like the processor-based system **100** in FIG. **1**) in a socket-to-socket connection. For example, the processor-based system **100** shown in FIG. **1** may be a first CPU motherboard system that can be communicatively coupled to another processor-based system through the internal communication network **114** and a coupled S2S interface circuit **122(0)-122(Y)**.

(31) Also as shown in FIG. **1**, the processor-based system **100** in this example also includes other interface (I/F) circuits **127(0)-127(Z)** as other examples of processing devices **110** that are also communicatively coupled to the internal communication network **114**. The interface circuits **127(0)-127(Z)** can provide an additional external communications interface to the SoC **102**, and

can be configured to provide a communication interface according to the desired standard or protocol. For example, the interface circuits **127(0)-127(Z)** could be PCIe interface circuits that are configured to support PCIe communications with the SoC **102**.

(32) Thus, in the processor-based system **100** in FIG. **1**, the internal communication network **114** enables different processing devices such as PU clusters **108(0)-108(N)** and their processor cores **112(0)-112(N)**, caches, the memory controllers **118(0)-118(M)**, the I/O interface circuits **120(0)-120(X)**, and/or the S2S interface circuits **122(0)-122(Y)** to work together efficiently. The fabric provided by the internal communication network **114** is designed to provide high bandwidth, low latency, and efficient routing of data between different processing devices of the processor-based system **100**.

(33) Also, as shown in FIG. **1** and as described in more detail below, the processor-based system **100** also includes a hierarchical power management system **124**. In this example, the hierarchical power management system **124** is integrated into the same IC chip **104** and in the same die **106** that includes the PU clusters **108(0)-108(N)** and the internal communication network **114**. The hierarchical power management system **124** is configured to control power consumption of the processor-based system **100** by controlling power consumption of some or all of the processing devices **110** in the IC chip **104**. The hierarchical power management system **124** is configured to control power consumption to achieve a desired performance within an overall power budget for the IC chip **104**. For example, the processor-based system **100** may have an overall power budget that is based on the ability of the IC chip **104** to dissipate heat generated by the operation of the processor-based system **100**. The processor-based system **100** may also have an overall power budget that is based on a current limit of power rails in the IC chip **104**. The power budget of the processor-based system **100** may also be based on power supply limits of a power supply that is powering the processor-based system **100**. Thus, the hierarchical power management system **124** can be configured to control power consumption by controlling the power level (e.g., voltage level, operating frequency) distributed at one or more of the power rails in the IC chip **104** that provide power to the processing devices **110**. For example, the hierarchical power management system **124** can be configured to cause additional power to be supplied to certain power rails thereby supplying power to higher current demanding devices to achieve higher performance, while providing less power to other power rails to keep the overall power within power and/or thermal limits for the IC chip **104**. For example, the hierarchical power management system **124** can be configured to communicate with or include a power management integrated circuit (PMIC) chip **125** (that can either be on-chip or off-chip to the SoC **102**) to actually cause the power supplied to certain power rails to be adjusted.

(34) Also, as discussed in more detail below, the hierarchical power management system **124** can also be configured to control power consumption in the processor-based system **100** by throttling performance (e.g., frequency and/or voltage) of the processing devices **110** in the processor-based system **100**. Throttling may refer to any measure (for example, modifying a clock frequency, and/or a supply voltage) to effect (i.e., reduce, maintain, or increase) power consumption. This in turn throttles (i.e., reduces, maintains, or increases) the current demand of such processing devices **110** and thus their power consumption in the IC chip **104**. Performance of clocked circuits in the processing devices **110** in the processor-based system **100** in terms of frequency (f) is related to power (P) according to the power equation $P = c f V_{sup}^2$, where ' c ' is capacitance and ' V ' is voltage. Thus, reducing frequency and/or voltage of a clocked circuit in a processing device **110** in the processor-based system **100** also reduces its power consumption.

(35) FIG. **2** is a logic diagram of the exemplary processor-based system **100** in FIG. **1** illustrating processing devices **110** communicatively coupled to the internal communication network **114**, and the hierarchical power management system **124** to control power consumption in the IC chip **104**. Common elements in the processor-based system **100** in FIGS. **1** and **2** are shown with common elements numbers and thus are not re-described. The IC chip **104** can also include target devices

200 whose control affects power, which can include the processing devices **110** and other circuits that are described below. As will be discussed in more detail below, the hierarchical power management system **124** is configured to throttle power to target devices **200** as well as processing devices **110** to throttle power consumption in the IC chip **104**.

(36) As also shown in FIG. 2, the hierarchical power management system **124** includes a centralized power estimation and limiting (PEL) circuit **126** that is configured to estimate power consumption in the IC chip **104** and take actions to limit or throttle power consumption in the IC chip **104**. In this example, the PEL circuit **126** can be provided as part of a power management integrated circuit (PMIC) **125** that is integrated in the IC chip **104**. The PEL circuit **126** may communicate such power throttling requests to a power management controller (PMC) **128**, for example, that is configured to control power provided by voltage rails in the IC chip **104**. Throttling power consumption can include both increasing power (e.g., increasing voltage to power rails) to increase power consumption for increased performance as well as decreasing power (e.g., decreasing voltage to power rails) to decrease power consumption. The hierarchical power management system **124** is configured to estimate power consumption in the IC chip **104** through receipt of power events **130** reported to it from devices at lower hierarchical levels in the IC chip **104** that provide information that provides an indirect indication of power consumption. For example, the IC chip **104** may have one or more temperature sensor(s) **132** that are configured to report thermal power events **130(1)** to the PEL circuit **126** to provide an indication of the temperature in the IC chip **104** which can then be correlated to power consumption by the processor-based system **100** in the IC chip **104**. As another example, the IC chip **104** may have one or more telemetry sensor(s) **134** (e.g., current sensors) that are configured to detect and report telemetry power events **130(2)** to the PEL circuit **126** to provide an indication of the telemetry information in the IC chip **104** which can then also be correlated to power consumption by the processor-based system **100** in the IC chip **104**.

(37) The power consumption of the processing devices **110** in the processor-based system **100** contributes to the power consumption in the IC chip **104**. Thus, it may be desired to also have a way for the PEL circuit **126** in the hierarchical power management system **124** to receive a direct indication of power consumption for the processing devices **110**. The PEL circuit **126** can then also use this information to estimate power consumption in the IC chip **104** and use such information to throttle the power consumption in the IC chip **104**. In this regard, as shown in FIG. 2, hierarchical power management system **124** also includes local area management (LAM) circuits **136** that are each associated with one or more processing devices **110** in the IC chip **104**. The LAM circuits **136** could be placed in various places in the IC chip **104**, including at corners of the IC chip **104** where power estimation and power limiting may need to be performed. For example, LAM circuits **136(1)** **(0)-136(1)(N)** may be associated with one or more of the PU clusters **108(0)-108(N)** as shown in FIG. 2. As another example, LAM circuits **136(2)-136(5)**, **136(6)(0)-136(6)(X)** may also be associated with respective one or more of the memory controllers **118**, the internal communication network **114**, e.g., the fabric, one or more of the I/O interface circuits **120**, the one or more of the S2S circuits **122**, and/or one or more interface circuits **127(0)-127(Z)**. Each LAM circuit **136(2)-136(5)**, **136(6)(0)-136(6)(X)** is configured to monitor the activity related to its associated processing device **110** as a monitored processing device **110** to then generate respective activity power events **138(1)(0)-138(1)(N)**, **138(2)-138(5)**, **138(6)(0)-138(6)(Z)** that are communicated directly or indirectly to the PEL circuit **126**. The activity power events **138(1)(0)-138(1)(N)**, **138(2)-138(5)**, **138(6)(0)-138(6)(Z)** contain information that relates to power consumption of the respective monitored processing device **110**. For example, the activity power events **138(1)(0)-138(1)(N)**, **138(2)-138(5)**, **138(6)(0)-138(6)(Z)** could contain processing activity information, or power consumption information that is generated by the respective LAM circuits **136(1)(0)-136(1)(N)**, **136(2)-136(5)**, **136(6)(0)-136(6)(X)** estimating power consumption of its monitored processing device **110** based on processing activity of its monitored processing device **110**. In

either case, in this manner, the activity power events **138** can be reported from local areas in the IC chip **104** where power estimations for particular monitored processing devices **110** are performed, to the centralized PEL circuit **126**. The PEL circuit **126** can then be configured to use the received activity power events **138** and/or the other power events **130** to estimate and control (i.e., throttle) power in the processor-based system **100** in the IC chip **104** to achieve a desired performance within an overall power budget for the IC chip **104**. For example, the activity power events **138** that are associated with estimations of power consumption of processing devices **110** that can be thought of in essence as power throttle recommendations to the PEL circuit **126** for the PEL circuit **126** to throttle power in the IC chip **104** if the estimated power consumption exceeds the power limits of the IC chip **104** or negatively affects performance in an undesired manner.

(38) By the PEL circuit **126** being configured to receive activity power events **138** relating to activity for individual processing devices **110** in the processor-based system **100**, this may also allow the PEL circuit **126** to throttle power consumption locally to certain processing devices **110** that are responsible for increased power consumption. This allows the PEL circuit **126** to throttle power with more discrimination rather than solely throttling power to power rails or in other ways in the IC chip **104** that affects the power delivered to a larger set of processing devices **110** as a whole. For example, as discussed in more detail below, the PEL circuit **126** can be configured to use the received activity power events **138** to perform performance throttling of processing devices **110** in the processor-based system **100** to throttle its power consumption. The PEL circuit **126** can be configured to generate power limiting management responses **140** to be communicated to certain LAM circuits **136** in the processor-based system **100** to cause such LAM circuits **136** to limit performance of its monitored processing device **110**.

(39) Performance throttling of a processing device **110** in the processor-based system **100** to throttle its power consumption can be accomplished in different manners. For example, as discussed in more detail below, performance throttling can be achieved by the PEL circuit **126** by generating a throughput throttling power limiting management response **140**, which is destined for the LAM circuit **136(3)** associated with the internal communication network **114**. The LAM circuit **136(3)** can be configured to throttle the throughput of communication traffic in the internal communication network **114**, such as at a particular network node in the internal communication network **114**, to throttle current demand in the internal communication network **114** and thus its power consumption. Throughput throttling can be isolated to only certain areas or network nodes in the internal communication network **114**. In another example, as discussed in more detail below, performance throttling in the processor-based system **100** can be achieved by the PEL circuit **126** by generating a clock throttling power limiting management response **140** to cause a clock circuit (which may be clocking one or more of the processing devices **110**) to throttle the speed (i.e., clock frequency) of certain clocked processing devices **110**. Clock throttling of a processing device **110** throttles its current demand which throttles its power consumption. In another example, as discussed in more detail below, performance throttling in the processor-based system **100** can be achieved by throttling or changing power states of a monitored processing device **110** to throttle its performance and thus its power consumption.

(40) FIG. 3 is a top view of an exemplary physical layout the semiconductor die (“die”) **102** of the IC chip **104** in FIG. 1 that includes the processor-based system **100** to illustrate further exemplary details of the physical layout of the hierarchical power management system **124** and an exemplary organization of power rails provided in the processor-based system **100**.

(41) As shown in FIG. 3, the IC chip **104** has a physical layout that includes a center tile CTILE, a west tile WTILE, an east tile ETILE, a south tile STILE, a north tile NTILE, and an A-tile ATILE. A tile is a smaller section of a semiconductor die that has been processed in a wafer process and contains a set of IC components. The center tile CTILE in this example includes the PU clusters **108(0)-108(N)**, shown as NCC0-NCC19. Different numbers of processor cores can be provided in different PU clusters **108(0)-108(N)**, NCC0-NCC19. In this example, the PU clusters

108(0)-108(N), **NCC0-NCC19** are all powered by a same power rail **300(1)**. The center tile CTILE in this example also includes the internal communication network **114**, which is shown by a plurality of center network nodes **FABC00-FABC65**. The network nodes **FABC00-FABC65** are circuits that create a network fabric (“fabric”) of communication paths between the different PU clusters **108(0)-108(N)** and the supporting processing devices **110**. In this example, the network nodes **FABC00-FABC65** are powered by a second power rail **300(2)**. The network nodes **FABC00-FABC65** are circuits that can include interconnected switches and/or routers, that provide a reliable network fabric that provides and receives data on the internal communications network **104** between different PU clusters **108(0)-108(N)** and the supporting processing devices **110**. The center tile CTILE in this example also includes the system level cache memory **116(0)-116(7)** powered by a third power rail **300(3)** to provide shared cache memory **116** for the PU clusters **108(0)-108(N)**, **NCC0-NCC19**. The system level cache memory **116(0)-116(7)** that is organized into different quadrants adjacent to and coupled to respective memory circuits **DDR0-DDR7** that include respective memory controllers **118(0)-118(7)** and coupled memory **304(0)-304(7)** (e.g., dynamic data random access memory (DDR) circuits) in the west tile WTILE to provide memory interlacing schemes for example. The memory circuits **DDR0-DDR7** may be powered by yet a separate, fourth power rail **300(4)**. The memory circuits **DDR0-DDR7** are also communicatively coupled to the internal communication network **114** through the respective network nodes **FABC00-FABC05**.

(42) With continuing reference to FIG. 3, the center tile CTILE in this example also includes the system level cache memory **116(8)-116(15)**, also powered by the third power rail **300(3)**, to provide additional shared cache memory **116** for the PU clusters **108(0)-108(N)**, **NCC0-NCC19**. The system level cache memory **116(8)-116(15)** may be organized into different quadrants adjacent to respective memory circuits **DDR8-DDR15** that include respective memory controllers **118(8)-118(15)** and coupled memory **304(8)-304(15)** (e.g., DDR circuits) in the east tile ETILE to provide memory interlacing schemes for example. The memory circuits **DDR8-DDR15** are also shown as being powered by the same fourth power rail **300(4)** as is powering the memory circuits **DDR0-DDR7** in the west tile WTILE. The memory circuits **DDR8-DDR15** are also communicatively coupled to the internal communication network **114** through the respective network nodes **FABC60-FABC65**.

(43) With continuing reference to FIG. 3, the center tile CTILE of the IC chip **104** in this example includes request node circuits **FABS00, FABS40, FABN57, FABN47** that are coupled to the internal communication network **114** to provide network interfaces between the I/O interface circuits **120(0)-120(3), 120(4)-120(7)** and the internal communication network **114** in the respective south tile STILE and north tile NTILE. The request node circuits **FABS00, FABS40, FABN57, FABN47** manage the traffic requests from the I/O interface circuits **120(0)-120(3), 120(4)-120(7)** to the internal communication network **114** and vice versa. The request node circuits **FABS00, FABS40, FABN57, FABN47** and the I/O interface circuits **120(0)-120(3), 120(4)-120(7)** in this example are powered by a fifth power rail **300(5)**.

(44) With continuing reference to FIG. 3, the A-tile ATILE in the IC chip **104** includes the PEL circuit **126** and the PMC **128** of the hierarchical power management system **124** in this example.

(45) Thus, as shown in FIG. 3, the processing devices **110** in the processor-based system **100** in the IC chip **104** are powered by a series of different power rails **300(1)-300(5)**. Thus, the PEL circuit **126** in the hierarchical power management system **124** has the resolution of each of these different power rails **300(1)-300(5)** in which to vary the voltage on such power rails **300(1)-300(5)** to throttle power consumption in the IC chip **104** based on the power events **130, 138**. Note that each power rail **300(1)-300(5)** can actually be included as a single or multiple power rails.

(46) FIG. 4 is a table **400** illustrating an exemplary assignment of power management circuits **AK0-AK5** in the PMIC **125** in the processor-based system **100**, to devices in the processor-based system **100** for supplying power to such devices. Power management circuits **AK0-AK5** can be responsible to control one or more different power rails **300(1)-300(5)** as shown in FIG. 3 to supply

power to various components. Multiple devices in the processor-based system **100** can be coupled to the same power rail **300(1)-300(5)** to receive power. For example, as shown in FIG. **4**, in this example, PU clusters NCC19, 18, 15, 14 are powered from power rails controlled by power management circuit AK0, PU clusters NCC 11-10 are powered from power rails controlled by power management circuit AK1, PU clusters NCC2, 3, 6, 7 are powered from power rails controlled by power management circuit AK2, PU clusters NCC0, 1, 4, 5 are powered from power rails controlled by power management circuit AK3, PU clusters NCC9-8 are powered from power rails controlled by power management circuit AK4, and PU clusters NCC12, 13, 16, 17 are powered from power rails controlled by power management circuit AK5. Also, as shown in FIG. **4**, a single device in the processor-based system **100** can be coupled to more than one power rail to receive power. For example, power supplied to the logic circuits (SoC_Logic) can be controlled by the multiple power management circuits AK1-AK4. The cache memory **116** can be supplied power from power rails controlled by the power management circuits AK0-AK5. Different memory controllers **118** are shown as being powered by power rails controlled by the power management circuits AK0-AK5. The I/O interface circuits **120(0)-120(3)** are shown as being powered by power rails controlled by separate respective power management circuits AK3, AK2, AK5, AK0.

(47) FIG. **5** is another top view of the processor-based system **100** in the IC chip **104** in FIG. **1** illustrating the local area management (LAM) circuits **136**, and the PEL circuit **126** as part of the hierarchical power management system **124**. As discussed above with regard to FIG. **2**, the LAM circuits **136** are configured to locally monitor activity of processing devices **110**, such as the PU clusters **108(0)-108(N)** in the processor-based system **100** to estimate and throttle its power consumption and report activity power events **138** regarding estimated power consumption to the PEL circuit **126**. The processor-based system **100** in this example includes a clock circuit **506** that generates a clock signal **508** to clock the PU clusters **108(0)-108(N)** to control the speed of the PU clusters **108(0)-108(N)**. The PEL circuit **126** is configured to collect activity power events **138** regarding power consumption of the monitored processing devices **110** and issue power limiting management responses **140** in response to throttle power consumption in the IC chip **104**.

(48) As shown in FIG. **5**, a plurality of LAM circuits **136(3)** are distributed in the center tile CTILE and associated with respective network node **500** (as processing devices **110**) of the internal communication network **114**.) For example, the internal communication network **114** can be a mesh network like shown in FIG. **5**. The internal communication network **114** is capable of routing communication traffic from the PU clusters **108(0)-108(N)** through different network nodes **500** based on performance and traffic characteristics of the internal communication network **114**. In this manner, the throughput of the internal communication network **114** is not limited by any single network node **500**. The processor-based system **100** in this example includes a clock circuit **510** that generates a clock signal **512** to clock the network nodes **500** to control the speed of the internal communication network **114**. The clock circuit **510** is another example of a target device **200** in the IC chip **104**. As will be discussed in more detail below, the LAM circuits **136(3)** associated with the network nodes **500** in the internal communication network **114** are configured to sample processing activity of respective assigned network nodes **500** to generate a plurality of activity samples. The LAM circuits **136(3)** are then configured to estimate power consumption of the assigned network node **500** based on the activity samples regarding its assigned network node **500** to generate an activity power event **138** based on such estimated power consumption of the respective network node **500**.

(49) Also, as shown in FIG. **5**, in this example, the hierarchical power management system **124** also includes regional activity management (RAM) circuits **502(3)** configured to monitor activity of the internal communication network **114**. The RAM circuits **502(3)** are located in a particular region of the internal communication network **114** with each being assigned and coupled to a subset of the LAM circuits **136(3)**. The RAM circuits **502(3)** are intermediate power management circuits in the hierarchical power management system **124**. The RAM circuits **502(3)** are coupled to the PEL

circuit **126** through a second communication network **504**. The RAM circuits **502(3)** are communicatively and hierarchically located between the LAM circuits **136(3)** and the centralized PEL circuit **126**. The RAM circuits **502(3)** are configured to receive and aggregate activity power events **138** reported by assigned LAM circuits **136(3)** regarding activity of their monitored network node **500**. The RAM circuits **502(3)** can then aggregate these activity power events **138** and report an aggregated activity power event to the PEL circuit **126** so that the PEL circuit **126** can determine how power consumption of network nodes **500** should be throttled to achieve a desired overall performance of the internal communication network **114** while also maintaining power consumption within desired limits. The PEL circuit **126** can communicate a power limiting management response **140** back to a given RAM circuit **502(3)** to perform throughput throttling of a given network node(s) **500** in response to the power consumption of a network node(s) **500** being determined to exceed desired limits. For example, as discussed in more detail below, the RAM circuit **502(3)** can be configured to throttle throughput of a given network node(s) **500** by selectively enabling and disabling communication traffic through the network node(s) **500**.

(50) Also, as shown in FIG. 5, in this example, a plurality of LAM circuits **136(2)** are distributed in the west tile WTILE and the east tile ETILE and associated with respective memory circuits DDR0-DDR7, DDR8-DDR15 (as processing devices **110**). As will also be discussed in more detail below, the LAM circuits **136(2)** associated with the memory circuits DDR0-DDR7, DDR8-DDR15 are configured to sample processing activity of respective assigned memory circuits DDR0-DDR7, DDR8-DDR15 to generate a plurality of activity samples. The LAM circuits **136(2)** are then configured to estimate power consumption of the assigned memory circuit DDR0-DDR7, DDR8-DDR15 based on the activity samples regarding their assigned network node **500** to generate an activity power event **138** based on such estimated power consumption of the respective memory circuits DDR0-DDR7, DDR8-DDR15.

(51) Also, as shown in FIG. 5, in this example, the hierarchical power management system **124** also includes RAM circuits **502(2)** configured to monitor activity of the memory circuits DDR0-DDR7, DDR8-DDR15. The RAM circuits **502(2)** are located in a particular region of the memory circuits DDR0-DDR7, DDR8-DDR15 with each being assigned and coupled to a subset of the LAM circuits **136(2)**. The RAM circuits **502(2)** are communicatively and hierarchically located between the LAM circuits **136(2)** and the centralized PEL circuit **126**. The RAM circuits **502(2)** are coupled to the PEL circuit **126** through the second communication network **504**. The RAM circuits **502(2)** are configured to receive and aggregate activity power events **138** reported by assigned LAM circuits **136(2)** regarding activity of their monitored memory circuits DDR0-DDR7, DDR8-DDR15. The RAM circuits **502(2)** can then aggregate these activity power events **138** and report an aggregated activity power event to the PEL circuit **126** so that the PEL circuit **126** can determine how power consumption of the memory circuits DDR0-DDR7, DDR8-DDR15 should be throttled to achieve a desired overall performance of the memory circuits DDR0-DDR7, DDR8-DDR15 while also maintaining power consumption within desired limits. The PEL circuit **126** can communicate a power limiting management response **140** back to a given RAM circuit **502(2)** to perform throughput and/or performance throttling of a given memory circuit(s) DDR0-DDR7, DDR8-DDR15 in response to the power consumption of a memory circuit DDR0-DDR7, DDR8-DDR15 being determined to exceed desired limits. For example, as discussed in more detail below, the RAM circuit **502(2)** can be configured to throttle throughput and/or performance of a given memory circuit(s) DDR0-DDR7, DDR8-DDR15 by selectively enabling and disabling memory access requests/responses to the memory circuits DDR0-DDR7, DDR8-DDR15.

(52) Also, as shown in FIG. 5, in this example, the hierarchical power management system **124** also includes RAM circuits **502(4)** configured to monitor activity of the I/O interface circuits **120(0)-120(7)**. The RAM circuits **502(4)** are located in a particular region of the I/O interface circuits **120(0)-120(7)** with each being assigned and coupled to a subset of the LAM circuits **136(4)** as shown. The RAM circuits **502(4)** are communicatively and hierarchically located between the

LAM circuits **136(4)** and the centralized PEL circuit **126**. The RAM circuits **502(4)** are coupled to the PEL circuit **126** through the second communication network **504**. The RAM circuits **502(4)** are configured to receive and aggregate activity power events **138** reported by assigned LAM circuits **136(4)** regarding activity of their monitored I/O interface circuits **120(0)-120(7)**. The RAM circuits **502(4)** can then aggregate these activity power events **138** and report an aggregated activity power event to the PEL circuit **126** so that the PEL circuit **126** can determine how power consumption of the I/O interface circuits **120(0)-120(7)** should be throttled to achieve a desired overall performance of the I/O interface circuits **120(0)-120(7)** while also maintaining power consumption within desired limits. The PEL circuit **126** can communicate a power limiting management response **140** back to a given RAM circuit **502(4)** to perform throughput and/or performance throttling of a given I/O interface circuit(s) **120(0)-120(7)** in response to the power consumption of an I/O interface circuit(s) **120(0)-120(7)** being determined to exceed desired limits. For example, as discussed in more detail below, the RAM circuit **502(4)** can be configured to throttle throughput and/or performance of a given I/O interface circuit(s) **120(0)-120(7)** by selectively enabling and disabling access requests/responses to the I/O interface circuit(s) **120(0)-120(7)**.

(53) As shown back in FIG. 2, LAM circuits **136(1)(0)-136(1)(N)** can also be associated with each PU cluster **108(0)-108(N)** in the processor-based system **100** to sample activity therein to estimate power consumption in a respective PU cluster **108(0)-108(N)**. The LAM circuits **136(1)(0)-136(1)(N)** can be configured to generate activity power events **138** that includes the estimated power consumptions in response to a RAM circuit **502**, which in turn aggregates such activity power events **138** to the PEL circuit **126**. The RAM circuits **502** assigned to the subset of LAM circuits **136(1)(0)-136(1)(N)** are coupled to the PEL circuit **126** through the second communication network **504**. The PEL circuit **126** can generate power limiting management responses **140** in response to throttle the performance of the PU clusters **108(0)-108(N)**.

(54) As also shown in FIG. 2, LAM circuits **136(5)** can also be associated with each S2S interface circuit **122(0)-122(Y)** in the processor-based system **100** to sample activity therein to estimate power consumption in a respective S2S interface circuit **122(0)-122(Y)**. The LAM circuits **136(5)** can be configured to generate activity power events **138** that includes the estimated power consumptions in response to a RAM circuit **502**, which in turn aggregates such activity power events **138** to the PEL circuit **126**. The RAM circuits **502** assigned to the subset of LAM circuits **136(5)** are coupled to the PEL circuit **126** through the second communication network **504**. The PEL circuit **126** can generate power limiting management responses **140** in response to throttle the performance of the S2S interface circuits **122(0)-122(Y)**.

(55) As shown back in FIG. 2, LAM circuits **136(6)(0)-136(6)(X)** can also be associated with each interface circuit **127(0)-127(Z)** in the processor-based system **100** to sample activity therein to estimate power consumption in a respective interface circuit **127(0)-127(Z)**. The LAM circuits **136(6)(0)-136(6)(X)** can be configured to generate activity power events **138** that includes the estimated power consumptions in response to a RAM circuit **502**, which in turn aggregates such activity power events **138** to the PEL circuit **126**. The RAM circuits **502** assigned to a subset of LAM circuits **136(6)(0)-136(6)(X)** is coupled to the PEL circuit **126** through the second communication network **504**. The PEL circuit **126** can generate power limiting management responses **140** in response to throttle the performance of the interface circuits **127(0)-127(Z)**.

(56) In this example, any of the RAM circuits **502**, **502(2)-504(4)** discussed above can also include circuitry to behave functionally as a LAM circuit for an assigned processing device **110**. In this regard, any of the RAM circuits **502**, **502(2)-504(4)** can also be configured to sample the processing activity of its respective assigned processing device **110** to generate a plurality of activity samples for such processing device **110**. Such RAM circuits **502**, **502(2)-504(4)** can be configured to estimate power consumption of its assigned processing device **110** based on the activity samples regarding its assigned processing device **110** to generate an aggregated activity power event based on such estimated power consumption of the respective processing device **110**.

and the other received activity power events **138** from its coupled LAM circuits **136(1)(0)-(1)(N)**, **136(2)-136(5)**, **136(6)(0)-136(6)(X)**.

(57) Note that in any of the above referenced examples, the RAM circuits **502** are optional for any of the monitored processing devices **110**, and their respective LAM circuits **136(1)-136(6)** can be configured to directly communicate activity power events **138** directly to the PEL circuit **126**.

(58) FIG. **6** is a schematic diagram illustrating additional exemplary detail of a three (3) level hierarchical power management system **624** that can be provided as the hierarchical power management system **124** in the processor-based system **100** in the IC chip **104** in FIGS. **1-3** and **5**. Common elements between the hierarchical power management system **624** in FIG. **6** and the hierarchical power management system **124** in FIGS. **1-3** and **5** are shown with common element numbers. In this regard, FIG. **6** illustrates a single LAM circuit **136** communicatively coupled to a single RAM circuit **502** which is coupled to the PEL circuit **126**. Note however that this is to simplify the illustration in FIG. **6**. In the hierarchical power management system **624** in FIG. **6**, there can be a plurality of RAM circuits **502** that are communicatively coupled to the PEL circuit **126**. There can also be a plurality of LAM circuits **136** that are communicatively coupled to each RAM circuit **502** of the plurality of RAM circuits **502**. The discussion below regarding the exemplary operation of the LAM circuit **136** and RAM circuit **502** are equally applicable to any number of LAM circuits and RAM circuits included in the processor-based system, including the LAM circuits **136(1)(0)-(1)(N)**, **136(2)-136(5)**, **136(6)(1)-136(6)(X)** and the RAM circuits **502**, **502(2)-502(4)**.

(59) With reference to FIG. **6**, the LAM circuit **136** in this example is configured to sample the processing activity as a received activity sample **600** of an assigned, monitored processing device **110** in each cycle of a given local time window. The LAM circuit **136** periodically samples activity of its monitored processing device **110** in a local time window representing the activity of the assigned, monitored processing device **110** in that local time window. In this example, the LAM circuit **136** is configured to correlate received activity samples **600** into a power consumption during a given local time window for the activity of the processing device **110** for that given local time window. The LAM circuit **136** includes an accumulate circuit **602** that is configured to accumulate the estimated power consumptions based on the received activity samples **600** sampled in a given local time window to generate an estimated current demand **604** for the monitored processing device **110** for the local time window. The estimated current demand **604** is an estimate of the accumulated current measurement reported by the assigned processing device **110** (i.e., power consumption) over the local time window. The accumulate circuit **602** then provides the estimated current demand **604** (current demand over time) for each local time window in a generated activity power event **606** on the second communication network **504** representing the estimated power consumption of the monitored processing device **110** that is communicated to the RAM circuit **502** assigned to the LAM circuit **136**. The accumulate circuit **602** repeats the same process for subsequent local time windows to accumulate the estimated power consumptions for received activity samples **600** during the local time window to generate a next estimated current demand **604** for the monitored processing device **110**.

(60) With continuing reference to FIG. **6**, the RAM circuit **502** includes an aggregation circuit **608** that is configured to aggregate the received activity power events **606** from its coupled LAM circuits **136** into a generated aggregated activity power event **138**. The RAM circuit **502** is then configured to communicate the aggregated activity power event **138** on the second communication network **504** to the PEL circuit **126**. Note that in this example, the RAM circuit **502** also includes its own LAM circuit **136R** that may be configured like the LAM circuit **136** in FIG. **6**. In this regard, the LAM circuit **136R** is configured to sample the processing activity **600R** of an assigned processing device **110** into a plurality of activity samples **600R**. The processing activity **600R** of the assigned processing device **110** is sampled periodically by the LAM circuit **136R** to generate a plurality of activity samples over a given local time window representing the activity of the

assigned, monitored processing device **110**. The LAM circuit **136R** is configured to determine a current flow rate and/or change in current flow rate (i.e., di/dt) of the assigned processing device **110** represented by the received plurality of activity samples **600R**. The LAM circuit **136R** can be programmed to correlate processing activity to power consumption to estimate the power consumption of the monitored processing device **110** over the local time window. The LAM circuit **136R** can then be configured to generate an activity power event **606R** representing the estimated power consumption of the monitored processing device **110** that is communicated to the aggregation circuit **608** of the RAM circuit **502** to be aggregated into the aggregated activity power event **138**.

(61) With continuing reference to FIG. 6, the PEL circuit **126** is configured to receive the aggregated activity power events **138** from the one or more RAM circuits **502** included in the hierarchical power management system **624**. In this example, the PEL circuit **126** includes a decode circuit **610** that is configured to decode the received aggregated activity power events **138** into a decoded activity power events **611** to be routed to a corresponding activity tracker circuit **612(1)-612(T)** that are each associated with a monitored processing device **110** in the processor-based system **100**. The PEL circuit **126** can also include other energy tracker circuits (not shown) that are associated with other power events (e.g., temperature, droop detection) that can also affect how the PEL circuit **126** decides to throttle power. The activity tracker circuits **612(1)-612(T)** are configured to aggregate associated activity power events **138** for an assigned monitored processing device **110** to determine whether power consumption for a monitored processing device **110** exceeds a defined threshold current flow rate/change in current flow rate. The activity tracker circuits **612(1)-612(T)** can also each include a power limit management policy that is configured to generate respective power throttle recommendations **614(1)-614(T)** for the PEL circuit **126** to use to determine how to throttle the distributed power and/or performance of the monitored processing devices **110** to throttle power consumption.

(62) With continuing reference to FIG. 6, the PEL circuit **126** also includes a merge circuit **616** that merges the power throttle recommendations **614(1)-614(T)** for the individual monitored processing devices **110** into merged power throttle recommendations **618(1)-618(Q)**. The merged power throttle recommendations **618(1)-618(Q)** are provided to respective assigned target circuits **620(1)-620(Q)**. Each target circuit **620(1)-620(Q)** is associated with a different target device **200** in the processor-based system **100** in which the PEL circuit **126** can issue power limiting management responses **140(1)-140(Q)** to limit the power consumption of such target device **200**. The target devices **200** are devices in the IC chip **104** whose operational control (e.g., operating voltage, frequency, workload) can affect power consumption in the IC chip **104**. The target devices in the IC chip **104** can include more than just the processing devices **110** in the processor-based system **100**. For example, the target devices **200** can include the power rails **300(1)-300(5)** as shown in FIG. 3 and/or any of the processing devices **110** in the processor-based system **100**. The PEL circuit **126** can be programmed to map (e.g., through firmware, electronic fuses, etc.) the merged power throttle recommendations **618(1)-618(Q)** to a particular target device **200**, and thus a target circuit **620(1)-620(Q)**, that may not directly correlate to each other. For example, it may be desired for the PEL circuit **126** to throttle power consumption of the I/O interface circuits **120(0)-120(X)** by not only throttling power consumption for the I/O interface circuits **120(0)-120(X)** but also by throttling power of the PU clusters **108(0)-108(N)** that may be contributing to the power consumption by the I/O interface circuits **120(0)-120(X)**. In this manner, the merged power throttle recommendations **618(1)-618(Q)** and/or other power events related to power issues and power consumption in the IC chip **104** can be mapped in the PEL circuit **126** to correlate to different target devices **200** for throttling power consumption. The merge circuit **616** can be programmed in a “many-to-many mapping” to correlate to different power limiting management responses within the IC chip **104** in the desired manner for more flexibility in managing power consumption in the IC chip **104** while still achieving the desired performance. In this manner, the power throttling

management behavior of the PEL circuit **126** can be configured and changed even after the IC chip **104** is deployed in an application.

(63) With continuing reference to FIG. **6**, the target circuits **620(1)-620(Q)** are each configured to determine if the power consumption of an associated target device **200** in the processor-based system **100** should be throttled based on the merged power throttle recommendations **618(1)-618(Q)** provided to the target circuits **620(1)-620(Q)**. The target circuits **620(1)-620(Q)** can each include finite state machine (FSM) circuits **622(1)-622(Q)** that are configured to analyze the respective received merged power throttle recommendation **618(1)-618(Q)** to determine if power consumption of an associated target device **200** should be throttled. If a FSM circuit **622(1)-622(Q)** determines that power consumption of an associated target device **200** in the processor-based system **100** should be throttled, the FSM circuit **622(1)-622(Q)** cause an associated power limiting command generation circuit **625(1)-625(Q)** to generate a power limiting management response **140(1)-140(Q)** to cause the power consumption of a target device **200** associated with the power limiting management response **140(1)-140(Q)** to limit power consumption.

(64) For example, if the target circuit **620(1)-620(Q)** is assigned to a target device **200** of a power rail **300(1)-300(5)**, the target circuit **620(1)-620(Q)** can be configured to determine how to throttle the voltage to the associated power rail **300(1)-300(5)** to control power consumption of processing devices **110** powered by such power rail **300(1)-300(5)**. The respective power limiting command generation circuit **625(1)-625(Q)** can be configured to generate a performance throttling power limiting management response **140(1)-140(Q)** to cause the voltage provided to the associated power rail **300(1)-300(5)** to be throttled to control power consumption of processing devices **110** powered by such associated power rail **300(1)-300(5)**.

(65) In another example, if the target circuit **620(1)-620(Q)** is assigned to a target device **200** such as the internal communication network **114**, the target circuit **620(1)-620(Q)** can be configured to determine how to throttle performance of the internal communication network **114** to control power consumption of the internal communication network **114**. For example, to throttle the throughput performance of the internal communication network **114**, the target device **200** may be the clock circuit **506** (FIG. **5**) that is configured to clock the internal communication network **114**. The clock circuit **506** is another example of a target device **200** in the IC chip **104**. The target circuit **620(1)-620(Q)** can determine a throttle frequency of the clock signal **508** generated by the clock circuit **506** (FIG. **5**) for generating a clock throttling power limiting management response **140(1)-140(Q)**. The clock throttling power limiting management response **140(1)-140(Q)** will cause the clock signal **508** to be throttled, which will in turn throttle the speed and the throughput performance of the internal communication network **114** and thus its power consumption and/or other circuits clocked by the clock signal **508**.

(66) In another example, if the target circuit **620(1)-620(Q)** is assigned to a target device **200** as a PU cluster **108(0)-108(N)** or any other processing device **110**, the target circuit **620(1)-620(Q)** can be configured to determine how to throttle performance of the internal communication network **114** to control power consumption of the internal communication network **114**. For example, to throttle performance of the PU cluster **108(0)-108(N)** or other processing device **110**, the target device **200** may also be the clock circuit **506** (FIG. **5**) that is configured to clock the PU clusters **108(0)-108(N)**. The target circuit **620(1)-620(Q)** can determine a throttle frequency of the clock signal **508** generated by the clock circuit **506** for generating a performance power limiting management response **140(1)-140(Q)**. The clock throttling power limiting management response **140(1)-140(Q)** will cause the clock signal **508** to be throttled, which will in turn throttle the performance of the PU clusters **108(0)-108(N)** or other processing devices **110**.

(67) As shown in FIG. **6**, in this example, to communicate the power limiting management responses **140(1)-140(Q)** generated by the PEL circuit **126** to effect a power throttling of a target device **200** in the processor-based system **100**, the power limiting management responses **140(1)-140(Q)** are communicated to a target device **200** in the processor-based system **100**. For

target devices **200** that are monitored processing devices **110** monitored by a LAM circuit **136** or RAM circuit **502**, the PEL circuit **126** can be configured to communicate an associated power limiting management response **140(1)-140(Q)** to the RAM circuit **502**. The RAM circuit **502** in this example includes a command processor **626** that is configured to receive a power limiting management response **140(1)-140(Q)** to process the power limiting management response **140(1)-140(Q)** to identify the LAM circuit **136** to communicate with to effectuate the power throttling requested in the received power limiting management response **140(1)-140(Q)**. In this example, the RAM circuit **502** includes a limiting command engine circuit **628** that is configured to generate a local power limiting management response **630** directed to the LAM circuit **136** that can effectuate the power throttling requested in the received power limiting management response **140(1)-140(Q)**. Note that if the local power limiting management response **630** is to throttle power consumption of multiple processing devices **110** monitored by multiple LAM circuits **136** associated with the RAM circuit **502**, the limiting command engine circuit **628** can address the local power limiting management response **630** to multiple LAM circuits **136**. Also note that in this example, if the RAM circuit **502** includes the LAM circuit **136R**, and the RAM circuit **502** is monitoring a processing device **110** that is the target device **200** to be throttled, the limiting command engine circuit **628** generates the local power limiting management response **630** directed to the LAM circuit **136R**.

(68) With continuing reference to FIG. **6**, in response to a LAM circuit **136** receiving a local power limiting management response **630**, a power limiting management decode and sequencer circuit **632** is configured to process the received local power limiting management response **630**. The power limiting management decode and sequencer circuit **632** is configured to determine the power throttling response to be effectuated to a monitored processing device **110** based on the local power limiting management response **630**. In this regard, the power limiting management decode and sequencer circuit **632** is configured to generate local throttle signals **634** to cause the power consumption in the processing device **110** to be throttled. For example, power limiting management decode and sequencer circuit **632** can be configured to generate a sequence of local throttle signals **634** to continually throttle up or down the power consumption of the monitored processing device **110** associated with its LAM circuit **136**.

(69) Note that in the sequence of operations and communications described above with regard to the LAM circuits **136** communicating activity power events **606** to the RAM circuits **502**, and the RAM circuits **502** communicating aggregated activity power events **138** to the PEL circuit **126**, communication delays are incurred. There is a delay between generating the activity samples **600** of sampling of power consumptions in a processing device **110** in a LAM circuit **136** and the reporting and receipt of an associated aggregated activity power event **138** in the PEL circuit **126**. This delay can be particularly large for an IC chip **104** that has a larger area, such as one that includes a number of PU clusters **108(0)-108(N)** and other processing devices **110** as in the processor-based system **100**. By the time the PEL circuit **126** receives the associated aggregated activity power event **138** and processes such to a generation of an associated power limiting management response **140(1)-140(Q)**, the power consumed by monitored processing device **110** may have already exceeded desired power limits in an undesired manner and/or for an undesired amount of time, possibly causing the power consumption in the IC chip **104** to exceed designed power limits. Further, instantaneous current demand by a monitored processing device **110** can cause di/dt events or voltage droop events that can cause performance issues and/or failures that may not be able to be timely addressed by the PEL circuit **126**.

(70) To mitigate the delay in the PEL circuit **126** receiving an aggregated activity power events **138** associated with monitored processing devices **110** in the processor-based system **100** that may affect throttling of power consumption within the processor-based system **100**, the LAM circuits **136**, **136R** can also be configured to directly throttle performance of its monitored processing device **110** to throttle its current demand and thus throttle its power consumption. This gives the

PEL circuit **126** more reaction time to receive and process aggregated activity power events **138** to determine how power consumption in the processor-based system **100** should be throttled to achieve a desired overall performance while also maintaining power consumption within desired limits. In this manner, then LAM circuits **136**, **136R** may be able to more timely mitigate a power issue by locally throttling power consumption of its specific monitored processing device **110** on a device granularity (without having to throttle performance in other processing devices **110**). The LAM circuits **136**, **136R** can be configured to continuously monitor and throttle power consumption locally in its monitored processing device **110** co-existent with the PEL circuit **126** generating power limiting management responses **140** to limit power consumption by target devices **200** in the processor-based system **100**.

(71) In this regard, as shown in FIG. **6**, the LAM circuit **136** in this example includes a di/dt circuit **636** to track the rate of change of power consumption by the processing device **110** for local power consumption throttling of its monitored processing device **110**. In this regard, the di/dt circuit **636** is configured to receive the estimated current demand **604** for the activity of the processing device **110** sampled by the LAM circuit **136** from the accumulate circuit **602** in each local time window. For each incoming estimated current demand **604** received (e.g., received for a given local time window), the di/dt circuit **636** is configured generate a next summed current demand **638** of such incoming estimated current demand **604** in the next local time window from the accumulate circuit **602** with one or more previous received estimated current demands **604** received for a previous estimated current demand **604** in a previous local time window. In this manner, the next summed current demand **638** is a running sum of the estimated current demands **604** for the processing device **110** over consecutive local time windows. The di/dt circuit **636** provides the next summed current demand **638** to an application processor **640** that provides a determined next current flow rate **642** based on the next summed current demand **638** to a throttle FSM circuit **644**. The throttle FSM circuit **644** is configured to determine on an ongoing basis whether the next current flow rate **642** of the assigned processing device **110** exceeds a threshold current flow rate or change in current flow rate configured for the monitored processing device **110** in the LAM circuit **136**. In response to determining that the next current flow rate **642** of the assigned processing device **110** exceeding the threshold current flow rate, the throttle FSM circuit **644** is configured to generate the local throttle signals **634** to throttle the power consumption of the monitored processing device **110**.

(72) In this manner, the LAM circuit **136** is configured to continually monitor the ongoing current flow rate of its monitored processing device **110** to be able to locally throttle the power consumption of the monitored processing device **110**. In this manner, the LAM circuit **136** is configured to more quickly respond to power consumption issues caused by the current demand of the monitored processing device **110**, such as di/dt events and voltage droops, before the PEL circuit **126** may be able to respond.

(73) As an example, if the monitored processing device **110** by the LAM circuit **136** is a network node **500** of the internal communication network **114**, the local throttle signals **634** generated by the LAM circuit **136** may be a throughput throttle to selectively enable and disable communication flow in the network node **500** to throttle its throughput thus throttling its power consumption. As another example, if the monitored processing device **110** by the LAM circuit **136** is a PU cluster **108(0)-108(N)** or other processing device **110**, the local throttle signals **634** generated by the LAM circuit **136** may be a performance throttle to selectively throttle performance or workload of the monitored PU cluster **108(0)-108(N)** or other processing device **110** to throttle its performance thus throttling its power consumption.

(74) Note that sampling of processing activity discussed herein may be accomplished by determining or sampling a quantity that is associated with an instantaneous activity of the monitored processing device **110**. For example, the workload performed by a monitored processing device **100** may be determined or discoverable as an indirect method to determine instantaneous activity that can be correlated to an estimated current or power consumption. As another example,

activity of a monitored processing device **110** may be determined by sensing a temperature at a temperature sensor associated with the processing device **110**. As another example, a voltage droop may be sensed at the processing device **110** to determine an activity sample. Also, other quantities may be used to sample activity. As an example, an incoming interrupt at the processing device, a status register, a state of an interrupt queue, or a signal indicating whether the processing device busy or idle, may be used for sampling of processing activity. In some examples, input source circuits such as LAM circuit **136** may determine input telemetry values based on samples of processing activity. In this respect, sampling of processing activity may be suitable for generating measurements for determining input telemetry values, such as input power telemetry values based on a measurement of one of capacitance, electric charge, energy, and current.

(75) Note that the components to perform local throttling by the LAM circuit **136** can also be provided in the LAM circuit **136R** in the RAM circuit **502** so that the LAM circuit **136R** is also configured to locally throttle a monitored processing device **110**.

(76) Note that the hierarchical power management system **124** provided in the IC chip **104** for the processor-based system **100** in FIG. **1** is not limited to the three (3) level hierarchical power management system **624** in FIG. **6**. For example, FIG. **7** is a schematic diagram of an alternative two (2) level hierarchical power management system **724** that can be provided as the hierarchical power management system **124** in the processor-based system **100** in the IC chip **104** in FIGS. **1-3** and **5**. The hierarchical power management system **724** in FIG. **7** is similar to the hierarchical power management system **624** in FIG. **6**, except that the intermediate RAM circuits **502** are not included in the hierarchical power management system **724** in FIG. **7**. The LAM circuits **136** are configured to provide activity power events **606** directly to the PEL circuit **126** to be processed. Common elements between the hierarchical power management system **724** in FIG. **7** and the hierarchical power management system **124** in FIGS. **1-3** and **5** are shown with common element numbers and are not re-described.

(77) Also, as discussed herein or the claims, it is stated that the PEL circuit **126** receives activity power events **606** from a LAM circuit **136**, this receipt of activity power events **606** can be directly from the LAM circuit **136** to the PEL circuit **126** or indirectly from one or more intermediate circuits, including the RAM circuits **502**. For example, as discussed above, the activity power events **606** generated by the LAM circuits **136** can be indirectly reported to the PEL circuit **126** the as part of being included in aggregated activity power events **138** generated and reported by a RAM circuit **502** to the PEL circuit **126** as part of received activity power events **606**.

(78) FIG. **8** is a flowchart illustrating an exemplary process **800** of the LAM circuits **136** and/or the RAM circuits **502** in the hierarchical power management systems **124**, **624**, **724** in FIGS. **1-3** and **5-7** locally monitoring and throttling power consumption of monitored processing devices **110**. The process **800** also includes the hierarchically reporting activity power events **606**, **138** related to the monitored power consumption by LAM circuits **136** and/or the RAM circuits **502** to throttle power consumption in the processor-based system **100** in response to the received activity power events **606**, **138**. The process **800** in FIG. **8** is discussed with regard to the hierarchical power management systems **624**, **724** as examples.

(79) In this regard, as shown in FIG. **8**, a first step of the process **800** can be sampling processing activity of an assigned processing device **110** of a plurality of processing devices **110** coupled to at least one power rails **300(1)-300(5)** of a plurality of power rails **300(1)-300(5)** to generate a plurality of activity samples **600** (block **802** in FIG. **8**). A next step in the process **800** can be determining a current flow rate **642** of the assigned processing device **110** based on the plurality of activity samples **600** (block **804** in FIG. **8**). A next step in the process **800** can be determining whether the current flow rate **642** of the assigned processing device **110** exceeds a defined threshold current flow rate (block **806** in FIG. **8**). A next step in the process **800** can be throttling the processing activity of the assigned processing device **110** to throttle its power consumption in response to determining the current flow rate **642** of the assigned processing device **110** exceeds

the threshold current flow rate (block **808** in FIG. **8**). Also, in addition to and/or in parallel to steps **804-808**, another step in the process **800** can be estimating power consumption of the assigned processing device **110** based on the plurality of activity samples **600** (block **810** in FIG. **8**). A next step in the process **800** can be generating an activity power event **606, 138** based on the estimated power consumption of the assigned processing device **110** (block **812** in FIG. **8**). A next step in the process **800** can be receiving a plurality of power events based on the activity power events **606, 138** (block **814** in FIG. **8**). A next step in the process **800** can be generating a power limiting management response **140** to cause power consumption to be throttled in the IC chip **104** based on the received plurality of activity power events **606, 138** (block **816** in FIG. **8**).

(80) FIG. **9A** is a schematic diagram illustrating exemplary detail of the di/dt circuit **636** and throttle FSM circuit **644** in the LAM circuit **136** shown in FIG. **6** to collect received estimated current demands **604** for processing activity of a monitored processing device **110** over local time windows and determine if a current flow rate and/or change in current flow rate of the monitored processing device **110** exceeds a threshold current flow rate. This information is used by the LAM circuit **136** to determine if its monitored processing device **110** should be locally throttled by its assigned LAM circuit **136** as previously discussed in FIG. **6**.

(81) In this regard, as shown in FIG. **9A**, the di/dt circuit **636** is configured to receive next estimated current demands **604** that are generated for each local time window of the LAM circuit **136** as discussed in FIG. **6**. The di/dt circuit **636** includes a plurality of latch circuits **900(1)-900(4)** that are clocked circuits (e.g., flip-flops) and are configured to store the incoming next estimated current demands **604** and previously received estimated current demands **604P(1)-604P(3)**. Latch circuit **900(1)** stores the next incoming estimated current demand **604**. The next incoming estimated current demand **604** stored in the latch circuit **900(1)** and the previous estimated current demands **604P(1)-604P(3)** stored in the latch circuits **900(1)-900(3)** are then shifted to the next respective latch circuit **900(2)-900(4)** for each newly received incoming estimated current demand **604** representing a local time window. For each incoming estimated current demand **604** received representing a local time window, the incoming estimated current demand **604** and previous estimated current demands **604P(1)-604P(3)** are provided to respective summing circuits **902(1)-902(4)**. The summing circuits **902(1)-902(3)** subtract the incoming estimated current demand **604** with a respective previous estimated current demand **604P(1)-604P(3)** to generate respective current flow rates over local time windows (i.e., change in current flow rates) **di_dt_1, di_dt_2, di_dt_3**, as discussed below, of the incoming estimated current demand **604** and the respective estimated current demands **604P(1)-604P(3)**. Thus, the determined change in current flow rates **di_dt_1, di_dt_2, di_dt_3** represent a rate in change in current flow rate or current demand and thus rate of change in power consumption of the monitored processing device **110** between the local time windows when the incoming estimated current demand **604** was received and a previous local time window of the respective previous estimated current demands **604P(1)-604P(3)**. **di_dt_1** is the change in current or current flow rate between respective estimated current demand **604** and **604P(1)**. **di_dt_2** is the change in current or current flow rate between respective estimated current demand **604** and **604P(2)**. **di_dt3** is the change in current or current flow rate between respective estimated current demand **604** and **604P(3)**.

(82) With continuing reference to FIG. **9A**, these change in current flow rates **di_dt_1, di_dt_2, di_dt_3** are then provided to a multiplexing circuit **904** that can selectively provide one of the change in current flow rates **di_dt_1, di_dt_2, di_dt_3** as the next current flow rate **642** to a comparator circuit **906** in the throttle FSM circuit **644**, discussed below. The selected change in current flow rate **di_dt_1, di_dt_2, di_dt_3** provided as the next current flow rate **642** to the multiplexing circuit **904** is based on a local time window selection signal **sel_di_dt_window** to select the local time windows to be compared to each current flow rate. This allows the flexibility of the di/dt circuit **636** to be programmed to select the local time windows of estimated current demands **604P(1), 604P(2)** to be compared to the incoming estimated current demand **604**. For

example, FIG. 9B is a graph 920 illustrating exemplary incoming and estimated current demands 604, 604P(1)-604P(3) collected by the di/dt circuit 636 in FIG. 9A plotted as a function of local time window to show how the incoming and estimated current demands 604, 604P(1)-604P(3) can be subtracted to generate respective change in current flow rates di_dt_1, di_dt_2, di_dt_3 between the incoming estimated current demand 604 and the estimated current demands 604P(1)-604P(3) over their respective local time windows twN, twN-1, twN-2, twN-3. The duration of the local time windows is known. Thus, the change in current flow rates di_dt_1, di_dt_2, di_dt_3 represent a change in current demand between the incoming estimated current demand 604 in a current local time window and a respective previous estimated current demand 604P(1)-604P(3) over the difference in their local time windows. The current flow rate curve 922 represents the current flow rate of a processing device 110 over a period of local time windows twN-3, twN-2, twN-1, and twN. As shown in FIG. 9B, the slope of the current flow rate curve 922 changes at each of the local time windows twN-3, twN-2, twN-1, and twN based on the change in current demand or change in current flow rate demanded of the processing device 110 between local time windows twN-3, twN-2, twN-1, and twN. FIG. 9B shows the basis on which the di/dt circuit 636 in FIG. 9A can generate the change in current flow rates di_dt_1, di_dt_2, di_dt_3 representing a change in current demand between the incoming estimated current demand 604 in a current local time window and a respective previous estimated current demand 604P(1)-604P(3) over the difference in their local time windows twN-3, twN-2, twN-1, and twN. This can be used to provide the current flow rate 642 of the processing device 110 to use to determine local power consumption throttling.

(83) The selected next current flow rate 642 is provided by the di/dt circuit 636 to the comparator circuit 906 in the throttle FSM circuit 644. The throttle FSM circuit 644 is configured to generate the local throttle signals 634 to throttle power consumption of the monitored processing device 110 based on whether the selected next current flow rate 642 (from selection of change in current flow rate di_dt_1, di_dt_2, di_dt_3) exceeds a threshold current flow rate (which can include a threshold change in current flow rate) for the monitored processing device 110. The threshold current flow rate for the monitored processing device 110 can be obtained from a current flow rate register 908. The current flow rate register 908 can be programmed with a threshold current flow rate for the monitored processing device 110. For example, the current flow rate register 908 can be programmed with different threshold current flow rates (e.g., lowest, level 1, level 2, highest) so that the comparator circuit 906 can generate local throttle signals 634 for different levels of power consumption throttling based on the comparison of selected next current flow rate 642 (from selection of change in current flow rate di_dt_1, di_dt_2, di_dt_3) with the selected threshold current flow rate obtained from the current flow rate register 908.

(84) Note that when current flow rate is discussed herein, such also means current flow and represents current (I) over a period of time (t) (I/t) or a change in the current flow rate (di/dt). A determined change in the current flow rate (di/dt) is determined from a determined current flow rate (t/T).

(85) The components of the hierarchical power management systems 124, 624, 724 in FIGS. 1-3 and 5-7 described above can be provided in different implementations. For example, FIG. 10 is a logic diagram of another exemplary PEL circuit 1026 that can be PEL circuit 126 provided in the hierarchical power management system 124, 624, 724 in FIGS. 1-3 and 5-7. Common elements between the PEL circuit 1026 in FIG. 10 and the PEL circuit 126 in the hierarchical power management system 124, 624, 724 in FIGS. 1-3 and 5-7 are shown with common element numbers.

(86) In this regard, as shown in FIG. 10, the PEL circuit 1026 is configured to receive the aggregated activity power events 138(1)-138(5) from the one or more RAM circuits 502. In this example, the PEL circuit 126 includes the decode circuit 610 that is configured to decode the received aggregated activity power events 138(1)-138(5) into the corresponding activity tracker circuit 612(1)-612(T) as previously described. The PEL circuit 1026 in this example also includes energy tracker circuits 1000(1)-1000(E) that are associated with energy power events 1002, such as

PMIC telemetry power events **1002(1)**, temperature events **1002(2)**, and voltage droop detection events **1002(3)** (all of which are examples of non-activity power events), that can also affect how the PEL circuit **126** decides to throttle power. The PEL circuit **1026** in this example also includes maximum average power (MAP) tracker circuits **1004(1)-1004(B)** that are circuit trackers that track the total power consumed in the SoC **102** according to a defined maximum power consumption limit. Similar to the activity tracker circuits **612(1)-612(T)**, the energy tracker circuits **1000(1)-1000(E)** and the MAP tracker circuits **1004(1)-1004(B)** are configured to respective energy power events **1002(1)-1002(3)** and/or aggregated activity power events **138(1)-138(5)** to determine whether a factor exists that is dependent on power consumption that exceeds a defined power (e.g., current) threshold/limit.

(87) The energy tracker circuits **1000(1)-1000(E)** each include respective data aggregator circuits **1016(1)-1016(E)** that are configured to aggregate the received energy power events **1002** into respective aggregated energy power events **1018(1)-1018(E)**. The activity tracker circuits **1000(1)-1000(E)** also each include respective data aggregator circuits **1020(1)-1020(T)** that are configured to aggregate received energy power events into respective aggregated energy power events **1022(1)-1022(T)**. The MAP tracker circuits **1004(1)-1004(B)** also each include respective data aggregator circuits **1024(1)-1024(B)** that are configured to aggregate received energy power events into respective aggregated MAP power events **1027(1)-1027(B)**. The energy tracker circuits **1000(1)-1000(E)**, the activity tracker circuits **612(1)-612(T)**, and the MAP tracker circuits **1004(1)-1004(B)** in this example each include a respective energy power limit management policy circuits **1006**, activity power limit management policy circuits **1008**, and MAP power limit management policy circuits **1010** that are configured to generate respective energy power throttle recommendations **1012**, activity power throttle recommendations **614**, and MAP power throttle recommendations **1014**. These generated respective energy power throttle recommendations **1012**, activity power throttle recommendations **614**, and MAP power throttle recommendations **1014** are based on the respective received aggregated energy power events **1018(1)-1018(E)**, aggregated activity power events, **1022(1)-1022(T)**, aggregated MAP power events **1027(1)-1027(B)** for the PEL circuit **126** to process to determine how to throttle power consumption in the IC chip **104**.

(88) With continuing reference to FIG. **10**, the energy tracker circuits **1000(1)-1000(E)**, the activity tracker circuits **612(1)-612(T)**, and the MAP tracker circuits **1004(1)-1004(B)** are configured to compare a power consumption indicated by the respective aggregated energy power events **1018(1)-1018(E)**, aggregated activity power events **1022(1)-1022(T)**, and aggregated MAP power events **1027(1)-1027(B)**, to the respective energy power limit management policy circuits **1006**, activity power limit management policy circuits **1008**, and MAP power limit management policy circuits **1010** energy tracker circuits **1000(1)-1000(E)**, the activity tracker circuits **612(1)-612(T)**, and the MAP tracker circuits **1004(1)-1004(B)**. The energy tracker circuits **1000(1)-1000(E)**, the activity tracker circuits **612(1)-612(T)**, and the MAP tracker circuits **1004(1)-1004(B)** are then configured to generate the respective energy power throttle recommendations **1012**, activity power throttle recommendations **614**, and MAP power throttle recommendations **1014** based on the comparison of the power consumptions indicated by the respective aggregated power events **1018(1)-1018(E)**, **1022(1)-1022(T)**, **1027(1)-1027(B)** to the respective power limit management policy circuits **1006**, **1008**, **1010**. For example, the energy power limiting management policy circuits **1006**, the activity power limit management policy circuits **1008**, and the MAP limiting management policy circuits **1010** may each have respective a threshold power consumption that is compared to the respective aggregated power events **1018(1)-1018(E)**, **1022(1)-1022(T)**, **1027(1)-1027(B)** to determine the respective power throttle recommendations **1012**, **614**, **1014**.

(89) With continuing reference to FIG. **10**, the PEL circuit **1026** also includes the merge circuit **616** that merges the energy power throttle recommendations **1012**, generate respective activity power throttle recommendations **614**, and MAP power throttle recommendations **1014** into merged power throttle recommendations **618(1)-618(6)**. The merged power throttle recommendations

618(1)-618(6) are provided to respective assigned target circuits **620(1)-620(6)**. Note that each merged power throttle recommendations **618(1)-618(6)** can be influenced by power throttle recommendations from each of the energy power throttle recommendations **1012**, generate respective activity power throttle recommendations **614**, and MAP power throttle recommendations **1014**. Each target circuit **620(1)-620(6)** is associated with a different target device **200** in the processor-based system **100** in which the PEL circuit **1026** can issue power limiting management responses **140(1)-140(6)** to limit the power consumption of such target device **200**.

(90) The target devices **200** can include the interface circuits **127(1)-127(Z)** that can be throttled by power limiting management responses **140(1)** communicated to a RAM circuit **502(6)** and/or LAM circuit **136(6)** configured to throttle power consumption in such interface circuits **127(1)-127(Z)**. The target devices **200** can include the PU clusters **108(0)-108(N)** that can be throttled by power limiting management responses **140(2)** communicated to a RAM circuit **502(1)** and/or LAM circuit **136(1)** configured to throttle power consumption in such PU clusters **108(0)-108(N)**. The target devices **200** can include the internal communication network **114** that can be throttled by power limiting management responses **140(3)** communicated to a RAM circuit **502(3)** and/or LAM circuit **136(3)** configured to throttle power consumption in such internal communication network **114**. The target devices **200** can include the memory controllers **118(0)-118(M)** that can be throttled by power limiting management responses **140(4)** communicated to a RAM circuit **502(2)** and/or LAM circuit **136(2)** configured to throttle power consumption in such memory controllers **118(0)-118(M)**. The target devices **200** can include the I/O interface circuits **120(0)-120(X)** that can be throttled by power limiting management responses **140(5)** communicated to a RAM circuit **502(4)** and/or LAM circuit **136(4)** configured to throttle power consumption in such I/O interface circuits **120(0)-120(X)**. The target devices **200** can include the S2S interface circuits **122(0)-122(Y)** that can be throttled by power limiting management responses **140(6)** communicated to a RAM circuit **502(5)** and/or LAM circuit **136(5)** configured to throttle power consumption in such S2S interface circuits **122(0)-122(Y)**.

(91) The merge circuit **616** in the PEL circuit **1026** can be programmed to map (e.g., through firmware, electronic fuses, etc.) merged power throttle recommendations **618(1)-618(6)** to a particular target device **200**, and thus a target circuit **620(1)-620(6)**, that may not directly correlate to each other. In this manner, the merged power throttle recommendations **618(1)-618(6)** related to power issues and power consumption in the IC chip **104** can be mapped in the PEL circuit **1026** to correlate to different target devices **200** for throttling power consumption. The merge circuit **616** can be programmed in a “many-to-many mapping” to correlate to different power limiting management responses within the IC chip **104** in the desired manner for more flexibility in managing power consumption in the IC chip **104** while still achieving the desired performance. In this manner, the power throttling management behavior of the PEL circuit **1026** can be configured and changed even after the IC chip **104** is deployed in an application.

(92) With continuing reference to FIG. 10, the target circuits **620(1)-620(6)** are each configured to determine if the power consumption of an associated target device **200** in the processor-based system **100** should be throttled based on the merged power throttle recommendations **618(1)-618(6)** provided to the target circuits **620(1)-620(6)**. The target circuits **620(1)-620(6)** are each configured to analyze the respective received merged power throttle recommendation **618(1)-618(6)** to determine if power consumption of an associated target device **200** should be throttled. If a target circuit **620(1)-620(Q)** determines that power consumption of an associated target device **200** in the processor-based system **100** should be throttled, the target circuit **620(1)-620(Q)** causes an associated power limiting management response **140(1)-140(6)** to be generated to be communicated to a respective RAM circuit **502(1)-502(6)** and/or LAM circuit **136(1)-136(6)** cause the power consumption of a target device **200** associated with the power limiting management response **140(1)-140(Q)** to limit power consumption.

(93) As discussed above with regard to FIG. 2, the LAM circuits **136** may be configured to locally

monitor activity of the processing devices **110** to estimate and throttle power consumption and to report activity power events **138** regarding estimated power consumption to the PEL circuit **126**. Each of the LAM circuits **136** may compute power telemetry values by calculating a weighted sum of respective activity power events **138** over an aggregation time window. The weighted sums from the LAM circuits **136** in some aspects may be aggregated by RAM circuits such as the RAM circuits **502**, and/or may be routed to PEL circuits such as the PEL circuit **126**, where they may be used to generate a power limiting management response such as the power limiting management response **140**. The LAM circuits **136** and the RAM circuits **502** thus may be collectively referred to herein as “input source circuits.”

(94) The power telemetry values reported to the PEL circuit **126** by the input source circuits may vary from one another based on factors such as dynamic activity, type of exercised logic cone, and realized functionality. Some telemetry values, such as power telemetry values representing power measurements, may be reported to the PEL circuit **126** as unitless quantities, while some may be reported as one of a number of various units. Since different input source circuits providing data to the PEL **126** may report using varied units, precisions, and dynamic ranges, the processor-based system **100** (e.g., PEL circuit **126**) must convert the data into a common range and unit before processing the data (e.g., by aggregating data from the input source circuits for estimation for longer time windows). The processor-based system **100** must also convert unitless quantities into, e.g., units such as joules (for energy) or coulombs (for electric charge), as appropriate.

(95) In this regard, FIG. **11** illustrates an exemplary aspect of an IC chip **1100** providing a processor-based system **1102** that comprises a PEL circuit **1104** configured to convert power telemetry values into common data formats for power estimation and throttling. In particular, aspects according to FIG. **11** convert and store power telemetry values without the need for multiplier circuits or divider circuits, thus reducing the complexity and improving the performance of the PEL circuit **1104**. In FIG. **11**, the PEL circuit **1104** corresponds in functionality to, e.g., the PEL circuit **126** discussed in greater detail above. The PEL circuit **1104** of FIG. **11** is configured to receive input power telemetry values **1106(0)-1106(1)** from respective input source circuits **1108(0)-1108(1)**. The input source circuits **1108(0)-1108(1)** each may comprise, e.g., a LAM circuit such as one of the LAM circuits **136**, a RAM circuit such as one of the RAM circuits **502**, a processing unit, or a PMIC such as the PMIC **125**. Each of input power telemetry value may be received by the PEL circuit **1104** as part of an activity power event such as one of the activity power events **138**, and may represent a measurement of one of capacitance, electric charge, energy, and current, as non-limiting examples.

(96) It is to be understood that, while FIG. **11** illustrates the IC chip **1100** as comprising the PEL circuit **1104**, aspects of the present disclosure are not limited to IC chips and processor-based systems that comprise PEL circuits. Furthermore, while FIG. **11** illustrates input telemetry values as input power telemetry values and the common format telemetry value as a common format power telemetry value, aspects of the present disclosure are not limited to telemetry values representing power measurements. Rather, the methods, apparatuses, and non-transitory computer-readable media of the present disclosure may be useful in any processor-based systems where values from different sources, having different units, and/or comprising unitless quantities may be stored and converted to a common format.

(97) In order to aggregate input power telemetry values such as the input power telemetry value **1106(1)**, the PEL circuit **1104** performs conversion operations as necessary to convert the input power telemetry values to a common format with respect to a unit value of a least significant bit of the converted value. In conventional approaches, the unit value of the least significant bit of the converted value is one (1), and converting each of the input power telemetry values **1106(0)-1106(1)** may involve multiplying or dividing the input power telemetry values **1106(0)-1106(1)** by a power of 10, which would require the use of a multiplier circuit and/or a divider circuit. However, in exemplary operation, the PEL circuit **1104** of FIG. **11** converts the

input power telemetry value **1106(0)** into a corresponding common format power telemetry value **1110** by performing a bit shift operation on the input power telemetry value **1106(0)**, effectively multiplying or dividing the input power telemetry value **1106(0)** by a power of two (2). The common format power telemetry value **1110** is then treated by the PEL circuit **1104** as though a least significant bit of the common format power telemetry value **1110** represents a unit value greater than one (1), based on a quotient of the power of two (2) and a corresponding power of 10. (98) For instance, assume for the sake of illustration only that the input power telemetry value **1106(0)** is to be converted into a common format comprising milli-units (e.g., millijoules, millicoulombs, or the like). Assume further that the input power telemetry value **1106(0)** is provided in micro-units, such that the conversion to the common format in conventional fashion would involve dividing the input power telemetry value **1106(0)** by 10^3 or 1,000. The PEL circuit in this example is configured to convert the input power telemetry value **1106(0)** into the common format power telemetry value **1110** by performing a rightward bit shift of the input power telemetry value by 10 bits (i.e., dividing the input power telemetry value **1106(0)** by 2^{10} or 1,024), with the unit value of a least significant bit of the resulting common format power telemetry value **1110** being 1.024 based on the quotient of $2^{10}/10^3$.

(99) In some examples, the input power telemetry value **1106(0)** may be provided in nano-units such that converting the input power telemetry value **1106(0)** into the common format would conventionally require dividing the input power telemetry value **1106(0)** by 10^6 or 1,000,000. The PEL circuit in such examples is configured to convert the input power telemetry value **1106(0)** into the common format power telemetry value **1110** by performing a rightward bit shift of the input power telemetry value **1106(0)** by 20 bits (i.e., dividing the input power telemetry value **1106(0)** by 2^{20} or 1,048,576), with the unit value of a least significant bit of the resulting common format power telemetry value **1110** being 1.048 based on the quotient of $2^{20}/10^6$.

(100) Some examples may provide that the input power telemetry value **1106(0)** is provided in pico-units, such that converting the input power telemetry value **1106(0)** into the common format in conventional fashion would require dividing the input power telemetry value **1106(0)** by 10^9 or 1,000,000,000. In such examples, the PEL circuit is configured to convert the input power telemetry value **1106(0)** into the common format power telemetry value **1110** by performing a rightward bit shift of the input power telemetry value **1106(0)** by 30 bits (i.e., dividing the input power telemetry value by 2^{30} or 1,073,741,824), with the unit value of a least significant bit of the resulting common format power telemetry value **1110** being 1.073 based on the quotient of $2^{30}/10^9$.

(101) Some input source circuits, such as the input source circuit **1108(1)** of FIG. 11, may be configured to provide an input power telemetry value such as the input power telemetry value **1106(1)** that is already in the common format. In such aspects, the PEL circuit **1104** may be configured to first determine whether a conversion of each of the input power telemetry values **1106(0)**-**1106(1)** into a common format power telemetry value such as the common format power telemetry value **1110** is required. If so (e.g., as with the input power telemetry value **1106(0)** of FIG. 11), the PEL circuit **1104** converts the input power telemetry value **1106(0)** into the common format power telemetry value **1110** responsive to determining that the conversion of the input power telemetry value **1106(0)** into the common format power telemetry value **1110** is required. If the PEL circuit **1104** determines that conversion is not necessary, as is the case with the input power telemetry value **1106(1)** of FIG. 11, the conversion operations are not performed.

(102) The PEL circuit **1104** then aggregates the common format power telemetry value **1110** (and/or any input power telemetry values that did not require conversion, such as the input power telemetry value **1106(1)** of FIG. 11) into an aggregated power telemetry value **1112**. This may be performed, for example, by summing the common format power telemetry value **1110** with a current value of the aggregated power telemetry value **1112** having the same unit value for its least

significant bit. The PEL circuit **1104** subsequently generates a power limiting management response **1114** to cause power consumption to be throttled in the IC chip **1100** based on the aggregated power telemetry value **1112**.

(103) Some aspects may provide that a firmware **1116** of the processor-based system **1102** may subsequently receive the aggregated power telemetry value **1112** from the PEL circuit **1104**. It is to be understood that, while FIG. **11** shows the firmware **1116** as an integral element of the PEL circuit **1104**, in some aspects the firmware **1116** may be an element of a different component of the processor-based system **1102**. Upon receiving the aggregated power telemetry value **1112**, the firmware **1116** may perform post-processing to convert the aggregated power telemetry value **1112** to a converted power telemetry value **1118**, wherein a unit value represented by a least significant bit of the converted power telemetry value **1118** is one (1). This may be accomplished in some aspects by multiplying the aggregated power telemetry value **1112** by the unit value of the aggregated power telemetry value **1112**. For example, if the unit value of a least significant bit of the aggregated power telemetry value **1112** is 1.024, the firmware **1116** may convert the aggregated power telemetry value **1112** to the converted power telemetry value **1118** by multiplying the aggregated power telemetry value **1112** by 1.024. The converted power telemetry value **1118** may then be used by the firmware **1116** for other post-processing operations and/or estimations, configuration of configuration state registers (CSRs), and the like.

(104) To illustrate exemplary operations for converting input power telemetry values into common data format values where a least significant bit represents a unit value greater than one (1), FIGS. **12A-12B** provide a flowchart showing exemplary operations **1200**. For the sake of clarity, elements of FIG. **11** are referenced in describing FIGS. **12A-12B**. It is to be understood that, in some aspects, operations shown in FIGS. **12A-12B** may be performed in an order other than that illustrated herein, and/or may be omitted.

(105) The exemplary operations **1200** begin in FIG. **12A** with the processor-based system **1102** of FIG. **11** (e.g., using the PEL circuit **1104** of FIG. **11**) receiving an input telemetry value (e.g., the input power telemetry value **1106(0)** of FIG. **11**) from an input source circuit of a processor-based system of an IC chip, such as the input source circuit **1108(0)** of the processor-based system **1102** of the IC chip **1100** of FIG. **11** (block **1202**). Note that the input power telemetry value **1106(0)** is referred to as such in describing the example illustrated in FIG. **11**, but is also referred to more generally as the “input telemetry value **1106(0)**” below in describing more general aspects. Some aspects may provide that the processor-based system **1102** may determine that a conversion of the input power telemetry value **1106(0)** into a common format power telemetry value (e.g., the common format power telemetry value **1110** of FIG. **11**) is required (block **1204**). It is to be understood that the common format power telemetry value **1110** is referred to as such in describing the example illustrated in FIG. **11**, but is also referred to more generally as the “common format telemetry value **1110**” below in describing more general aspects.

(106) The processor-based system **1102** converts the input telemetry value **1106(0)** into the common format telemetry value **1110**, wherein a first unit value represented by a least significant bit of the common format telemetry value **1110** is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10 (block **1206**). In the aspects mentioned above and illustrated in FIG. **11**, the operations of block **1206** for converting the input power telemetry value **1106(0)** into the common format power telemetry value **1110** is performed responsive to determining that the conversion of the input power telemetry value **1106(0)** into the common format power telemetry value **1110** is required (block **1208**).

(107) According to some aspects, the operations of the operations of block **1206** for converting the input telemetry value **1106(0)** into the common format telemetry value **1110** may comprise the processor-based system **1102** performing a rightward bit shift of the input telemetry value **1106(0)** by 10 bits, wherein the first unit value is 1.024 based on the quotient of $2^{\text{sup.}10}/10^{\text{sup.}3}$ (block **1210**). Some aspects may provide that the operations of the operations of block **1206** for converting

the input telemetry value **1106(0)** into the common format telemetry value **1110** comprise the processor-based system **1102** performing a rightward bit shift of the input telemetry value **1106(0)** by 20 bits, wherein the first unit value is 1.048 based on the quotient of $2^{\text{sup.}20}/10^{\text{sup.}6}$ (block **1212**). In some aspects, the operations of the operations of block **1206** for converting the input telemetry value **1106(0)** into the common format telemetry value **1110** may comprise the processor-based system **1102** performing a rightward bit shift of the input telemetry value **1106(0)** by 30 bits, wherein the first unit value is 1.073 based on the quotient of $2^{\text{sup.}30}/10^{\text{sup.}9}$ (block **1214**). The exemplary operations **1200** then continue at block **1216** of FIG. **12B**.

(108) Turning now to FIG. **12B**, the exemplary operations **1200** continue with the processor-based system **1102** processing the common format telemetry value **1110**. In aspects in which the common format telemetry value **1110** comprises the common format power telemetry value **1110**, the operations of block **1216** for processing the common format telemetry value **1110** may comprise aggregating the common format power telemetry value **1110** into an aggregated power telemetry value such as the aggregated power telemetry value **1112** of FIG. **11** (block **1218**). The processor-based system **1102** then generates a power limiting management response (e.g., the power limiting management response **1114** of FIG. **11**) to cause power consumption to be throttled in the IC chip **1100** based on the aggregated power telemetry value **1112** (block **1220**).

(109) In some aspects illustrated in FIG. **11**, the processor-based system **1102** (e.g., using the firmware **1116**) may receive the aggregated power telemetry value **1112** (block **1222**). The processor-based system **1102** may convert the aggregated power telemetry value **1112** to a converted power telemetry value (e.g., the converted power telemetry value **1118** of FIG. **11**), wherein a second unit value represented by a least significant bit of the converted power telemetry value **1118** is one (1) (block **1224**). Some aspects may provide that the operations of block **1222** for converting the aggregated power telemetry value **1112** to the converted power telemetry value **1118** may comprise multiplying the aggregated power telemetry value **1112** by the first unit value (block **1226**).

(110) A hierarchical power management system that can be provided in an IC chip for an integrated processor-based system that is configured to locally monitor activity of devices in the processor-based system to locally estimate and throttle its power consumption, and report activity power events regarding estimated power consumption to a centralized PEL circuit configured to collect activity power events regarding power consumption of the monitored processing devices and throttle power in the IC chip in response, including but not limited to the hierarchical power management systems and their exemplary components in FIGS. **1-3**, **5-7**, and **9A-11**, and operating according to the exemplary process **800** in FIG. **8**, and according to any aspects disclosed herein. may be provided in or integrated into any processor-based device. Examples, without limitation, include a set top box, an entertainment unit, a navigation device, a communications device, a fixed location data unit, a mobile location data unit, a global positioning system (GPS) device, a mobile phone, a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a tablet, a phablet, a server, a computer, a portable computer, a mobile computing device, laptop computer, a wearable computing device (e.g., a smart watch, a health or fitness tracker, eyewear, etc.), a desktop computer, a personal digital assistant (PDA), a monitor, a computer monitor, a television, a tuner, a radio, a satellite radio, a music player, a digital music player, a portable music player, a digital video player, a video player, a digital video disc (DVD) player, a portable digital video player, an automobile, a vehicle component, an avionics system, a drone, and a multicopter.

(111) FIG. **13** is a block diagram of another exemplary processor-based system that includes a hierarchical power management system configured to locally monitor activity of devices in the processor-based system to locally estimate and throttle its power consumption, and report activity power events regarding estimated power consumption to a centralized PEL circuit configured to collect activity power events regarding power consumption of the monitored processing devices and throttle power in the IC chip in response, including but not limited to the hierarchical power

management systems **124**, **624**, **724** and their exemplary components in FIGS. **1-3**, **5-7**, and **9A-10**. (112) In this example, the processor-based system **1300** may be formed in an IC chip **1302** and as a system-on-a-chip (SoC) **1304**. The processor-based system **1300** includes a central processing unit (CPU) (s) **1306** that includes one or more processors **1308**, which may also be referred to as CPU cores or processor cores. The CPU **1306** may have cache memory **1310** coupled to the CPU **1306** for rapid access to temporarily stored data. The CPU **1306** is coupled to a system bus **1312** and can intercouple master and slave devices included in the processor-based system **1300**. As is well known, the CPU **1306** communicates with these other devices by exchanging address, control, and data information over the system bus **1312**. For example, the CPU **1306** can communicate bus transaction requests to a memory controller **1314**, as an example of a slave device. Although not illustrated in FIG. **13**, multiple system buses **1312** could be provided, wherein each system bus **1312** constitutes a different fabric.

(113) Other master and slave devices can be connected to the system bus **1312**. As illustrated in FIG. **13**, these devices can include a memory system **1316** that includes the memory controller **1314** and a memory array(s) **1318**, one or more input devices **1320**, one or more output devices **1322**, one or more network interface devices **1324**, and one or more display controllers **1326**, as examples. The input device(s) **1320** can include any type of input device, including, but not limited to, input keys, switches, voice processors, etc. The output device(s) **1322** can include any type of output device, including, but not limited to, audio, video, other visual indicators, etc. The network interface device(s) **1324** can be any device configured to allow exchange of data to and from a network **1328**. The network **1328** can be any type of network, including, but not limited to, a wired or wireless network, a private or public network, a local area network (LAN), a wireless local area network (WLAN), a wide area network (WAN), a BLUETOOTH™ network, and the Internet. The network interface device(s) **1324** can be configured to support any type of communications protocol desired.

(114) The CPU **1306** may also be configured to access the display controller(s) **126** over the system bus **1312** to control information sent to one or more displays **1330**. The display controller(s) **1326** sends information to the display(s) **1330** to be displayed via one or more video processor(s) **1332**, which process the information to be displayed into a format suitable for the display(s) **1330**. The display(s) **1330** can include any type of display, including, but not limited to, a cathode ray tube (CRT), a liquid crystal display (LCD), a plasma display, a light emitting diode (LED) display, etc. The IC chip **1302** also includes a PMIC **1334** that includes a PEL circuit **1336** as part of a hierarchical power management system **1338**. The PEL circuit **1336** can be the PEL circuit **126** in the hierarchical power management systems **124**, **624**, **724** in FIGS. **1-3**, **5-7**, and **9** as examples. The hierarchical power management system **1338** can include one or more LAM circuits **1340(1)-1340(6)** that are associated with one or more of the processors **1308**, the cache memory **1310**, the memory controller **1314**, the network interface device(s) **1324**, the display controller **1326**, and/or the system bus **1312** that are configured to monitor activity associated with these processing devices and reporting activity power events regarding activity of these devices within the hierarchical power management system **1338**. The LAM circuits **1340(1)-1340(6)** may be the LAM circuits **136**, **136R** in the hierarchical power management systems **124**, **624**, **724** in FIGS. **1-3**, **5-7**, and **10A** as examples. One or more RAM circuits **1342** may also be provided as part of the hierarchical power management system **1338** to receive activity power events from groupings of LAM circuits **1340(1)-1340(6)** to aggregate such activity power events into aggregated activity power events to be communicated to the PEL circuit **1336**. The RAM circuits **1342** may be the RAM circuits **502** in the hierarchical power management systems **124**, **624**, **724** in FIGS. **1-3**, and **5-7** as examples.

(115) FIG. **14** illustrates an exemplary wireless communications device **1400** that can include a hierarchical power management system **1402** configured to locally monitor activity of devices in the processor-based system to locally estimate and throttle its power consumption, and report

activity power events regarding estimated power consumption to a centralized PEL circuit configured to collect activity power events regarding power consumption of the monitored processing devices and throttle power in the IC chip in response, including but not limited to the hierarchical power management systems **124**, **624**, **724** and their exemplary components in FIGS. **1-3**, **5-7**, and **9A-11**.

(116) As shown in FIG. **14**, the wireless communications device **1400** includes a RF transceiver **1404** and a data processor **1406**. The RF transceiver **1404** and/or the data processor **1406** can include respective hierarchical power management systems **1402(1)**, **1402(2)** configured to locally monitor activity of devices in the processor-based system to locally estimate and throttle its power consumption, and report activity power events regarding estimated power consumption to a centralized PEL circuit configured to collect activity power events regarding power consumption of the monitored processing devices and throttle power in the IC chip in response, including but not limited to the hierarchical power management systems **124**, **624**, **724** and their exemplary components in FIGS. **1-3**, **5-7**, and **9A-10**.

(117) The components of the RF transceiver **1404** and/or data processor **1406** can be split among multiple different die **1403(1)**, **1403(2)**. The data processor **1406** may include a memory to store data and program codes. The RF transceiver **1404** includes a transmitter **1408** and a receiver **1410** that support bi-directional communications. In general, the wireless communications device **1400** may include any number of transmitters **1408** and/or receivers **1410** for any number of communication systems and frequency bands. All or a portion of the RF transceiver **1404** may be implemented on one or more analog ICs, RF ICs, mixed-signal ICs, etc.

(118) The transmitter **1408** or the receiver **1410** may be implemented with a super-heterodyne architecture or a direct-conversion architecture. In the super-heterodyne architecture, a signal is frequency-converted between RF and baseband in multiple stages, e.g., from RF to an intermediate frequency (IF) in one stage, and then from IF to baseband in another stage for the receiver **1410**. In the direct-conversion architecture, a signal is frequency-converted between RF and baseband in one stage. The super-heterodyne and direct-conversion architectures may use different circuit blocks and/or have different requirements. In the wireless communications device **1400** in FIG. **14**, the transmitter **1408** and the receiver **1410** are implemented with the direct-conversion architecture.

(119) In the transmit path, the data processor **1406** processes data to be transmitted and provides I and Q analog output signals to the transmitter **1408**. In the exemplary wireless communications device **1400**, the data processor **1406** includes digital-to-analog converters (DACs) **1412(1)**, **1412(2)** for converting digital signals generated by the data processor **1406** into the I and Q analog output signals, e.g., I and Q output currents, for further processing.

(120) Within the transmitter **1408**, lowpass filters **1414(1)**, **1414(2)** filter the I and Q analog output signals, respectively, to remove undesired signals caused by the prior digital-to-analog conversion. Amplifiers (AMPs) **1416(1)**, **1416(2)** amplify the signals from the lowpass filters **1414(1)**, **1414(2)**, respectively, and provide I and Q baseband signals. An upconverter **1418** upconverts the I and Q baseband signals with I and Q transmit (TX) local oscillator (LO) signals through mixers **1420(1)**, **1420(2)** from a TX LO signal generator **1422** to provide an upconverted signal **1424**. A filter **1426** filters the upconverted signal **1424** to remove undesired signals caused by the frequency upconversion as well as noise in a receive frequency band. A power amplifier (PA) **1428** amplifies the upconverted signal **1424** from the filter **1426** to obtain the desired output power level and provides a transmit RF signal. The transmit RF signal is routed through a duplexer or switch **1430** and transmitted via an antenna **1432**.

(121) In the receive path, the antenna **1432** receives signals transmitted by base stations and provides a received RF signal, which is routed through the duplexer or switch **1430** and provided to a low noise amplifier (LNA) **1434**. The duplexer or switch **1430** is designed to operate with a specific receive (RX)-to-TX duplexer frequency separation, such that RX signals are isolated from TX signals. The received RF signal is amplified by the LNA **1434** and filtered by a filter **1436** to

obtain a desired RF input signal. Downconversion mixers **1438(1)**, **1438(2)** mix the output of the filter **1436** with I and Q RX LO signals (i.e., LO_I and LO_Q) from an RX LO signal generator **1440** to generate I and Q baseband signals. The I and Q baseband signals are amplified by AMPs **1442(1)**, **1442(2)** and further filtered by lowpass filters **1444(1)**, **1444(2)** to obtain I and Q analog input signals, which are provided to the data processor **1406**. In this example, the data processor **1406** includes analog-to-digital converters (ADCs) **1446(1)**, **1446(2)** for converting the analog input signals into digital signals to be further processed by the data processor **1406**.

(122) In the wireless communications device **1400** of FIG. **14**, the TX LO signal generator **1422** generates the I and Q TX LO signals used for frequency upconversion, while the RX LO signal generator **1440** generates the I and Q RX LO signals used for frequency downconversion. Each LO signal is a periodic signal with a particular fundamental frequency. A TX phase-locked loop (PLL) circuit **1448** receives timing information from the data processor **1406** and generates a control signal used to adjust the frequency and/or phase of the TX LO signals from the TX LO signal generator **1422**. Similarly, an RX PLL circuit **1450** receives timing information from the data processor **1406** and generates a control signal used to adjust the frequency and/or phase of the RX LO signals from the RX LO signal generator **1440**.

(123) Those of skill in the art will further appreciate that the various illustrative logical blocks, modules, circuits, and algorithms described in connection with the aspects disclosed herein may be implemented as electronic hardware, instructions stored in memory or in another computer readable medium and executed by a processor or other processing device, or combinations of both. Memory disclosed herein may be any type and size of memory and may be configured to store any type of information desired. To clearly illustrate this interchangeability, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. How such functionality is implemented depends upon the particular application, design choices, and/or design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

(124) The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(125) The aspects disclosed herein may be embodied in hardware and in instructions that are stored in hardware, and may reside, for example, in Random Access Memory, flash memory, Read Only Memory (ROM), Electrically Programmable ROM (EPROM), Electrically Erasable Programmable ROM (EEPROM), registers, a hard disk, a removable disk, a CD-ROM, or any other form of computer readable medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a remote station. In the alternative, the processor and the storage medium may reside as discrete components in a remote station, base station, or server.

(126) It is also noted that the operational steps described in any of the exemplary aspects herein are described to provide examples and discussion. The operations described may be performed in

numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary aspects may be combined. It is to be understood that the operational steps illustrated in the flowchart diagrams may be subject to numerous different modifications as will be readily apparent to one of skill in the art. Those of skill in the art will also understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(127) The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations. Thus, the disclosure is not intended to be limited to the examples and designs described herein, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

(128) Implementation examples are described in the following numbered clauses:

(129) 1. An integrated circuit (IC) chip comprising a processor-based system, the processor-based system configured to:

(130) receive, from an input source circuit of the processor-based system, an input telemetry value; convert the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10; and process the common format telemetry value.

2. The IC chip of clause 1, wherein: the input telemetry value comprises an input power telemetry value; and the common format telemetry value comprises a common format power telemetry value; the input power telemetry value and the common format power telemetry value each represent a measurement of one of capacitance, electric charge, energy, and current.

3. The IC chip of clause 2, wherein the processor-based system is further configured to process the common format telemetry value by being configured to: aggregate the common format power telemetry value into an aggregated power telemetry value; and generate a power limiting management response to cause power consumption to be throttled in the IC chip based on the aggregated power telemetry value.

4. The IC chip of clause 3, wherein the processor-based system is further configured to: receive the aggregated power telemetry value; and convert the aggregated power telemetry value to a converted power telemetry value, wherein a second unit value represented by a least significant bit of the converted power telemetry value is one (1).

5. The IC chip of clause 4, wherein the IC chip is configured to convert the aggregated power telemetry value to the converted power telemetry value by being configured to multiply the aggregated power telemetry value by the first unit value.

6. The IC chip of any one of clauses 1-5, wherein: the IC chip is configured to convert the input telemetry value into the common format telemetry value by being configured to perform a rightward bit shift of the input telemetry value by 10 bits; and the first unit value is 1.024 based on the quotient of $2^{\text{sup.}10}/10^{\text{sup.}3}$.

7. The IC chip of any one of clauses 1-5, wherein: the IC chip is configured to convert the input telemetry value into the common format telemetry value by being configured to perform a rightward bit shift of the input telemetry value by 20 bits; and the first unit value is 1.048 based on the quotient of $2^{\text{sup.}20}/10^{\text{sup.}6}$.

8. The IC chip of any one of clauses 1-5, wherein: the IC chip is configured to convert the input telemetry value into the common format telemetry value by being configured to perform a

rightward bit shift of the input telemetry value by 30 bits; and the first unit value is 1.073 based on the quotient of $2^{\sup.30}/10^{\sup.9}$.

9. The IC chip of any one of clauses 1-8, wherein: the IC chip is further configured to determine whether a conversion of the input telemetry value into the common format telemetry value is required; and the IC chip is configured to convert the input telemetry value into the common format telemetry value responsive to determining that the conversion of the input telemetry value into the common format telemetry value is required.

10. The IC chip of any one of clauses 1-9, wherein: the IC chip comprises a power estimation and limiting (PEL) circuit; and the input source circuit comprises one of a local activity management (LAM) circuit of the IC, a regional activity management (RAM) circuit of the IC, a processing unit of the IC, and a power management IC (PMIC) of the IC.

11. The IC chip of clause 1, integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; a drone; and a multicopter.

12. An integrated circuit (IC) chip, comprising: means for receiving, from an input source circuit, an input telemetry value; means for converting the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10; means for processing the common format telemetry value.

13. A method for converting a telemetry value into a common data format in a processor-based system in an integrated circuit (IC) chip, the method comprising: receiving, from an input source circuit of the processor-based system, an input telemetry value; converting the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10; processing the common format telemetry value.

14. The method of clause 13, wherein: the input telemetry value comprises an input power telemetry value; the common format telemetry value comprises a common format power telemetry value; and the input power telemetry value and the common format power telemetry value each represent a measurement of one of capacitance, electric charge, energy, and current.

15. The method of clause 14, wherein processing the common format telemetry value comprises: aggregating the common format power telemetry value into an aggregated power telemetry value; and generating a power limiting management response to cause power consumption to be throttled in the IC chip based on the aggregated power telemetry value.

16. The method of clause 15, further comprising: receiving the aggregated power telemetry value; and converting the aggregated power telemetry value to a converted power telemetry value, wherein a second unit value represented by a least significant bit of the converted power telemetry value is one (1).

17. The method of clause 16, wherein converting the aggregated power telemetry value to the converted power telemetry value comprises multiplying the aggregated power telemetry value by the first unit value.

18. The method of any one of clauses 13-17, wherein: converting the input telemetry value into the common format telemetry value comprises performing a rightward bit shift of the input telemetry value by 10 bits; and the first unit value is 1.024 based on the quotient of $2^{\sup.10}/10^{\sup.3}$.

19. The method of any one of clauses 13-17, wherein: converting the input telemetry value into the common format telemetry value comprises performing a rightward bit shift of the input telemetry value by 20 bits; and the first unit value is 1.048 based on the quotient of $2^{\text{sup.20}}/10^{\text{sup.6}}$.
20. The method of any one of clauses 13-17, wherein: converting the input telemetry value into the common format telemetry value comprises performing a rightward bit shift of the input telemetry value by 30 bits; and the first unit value is 1.073 based on the quotient of $2^{\text{sup.30}}/10^{\text{sup.9}}$.
21. The method of any one of clauses 13-20, further comprising determining that a conversion of the input telemetry value into the common format telemetry value is required; wherein converting the input telemetry value into the common format telemetry value is responsive to determining that the conversion of the input telemetry value into the common format telemetry value is required.
22. The method of any one of clauses 13-21, wherein: the IC chip comprises a power estimation and limiting (PEL) circuit; and the input source circuit comprises one of a local activity management (LAM) circuit of the IC, a regional activity management (RAM) circuit of the IC, a processing unit of the IC, and a power management IC (PMIC) of the IC.
23. A non-transitory computer-readable medium, having stored thereon computer-executable instructions that, when executed, cause a processor of a processor-based device to: receive, from an input source circuit of the processor-based system, an input telemetry value; convert the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10; process the common format telemetry value.
24. The non-transitory computer-readable medium of clause 23, wherein: the input telemetry value comprises an input power telemetry value; and the common format telemetry value comprises a common format power telemetry value; the input power telemetry value and the common format power telemetry value each represent a measurement of one of capacitance, electric charge, energy, and current.
25. The non-transitory computer-readable medium of clause 24, wherein the computer-executable instructions cause the processor to process the common format telemetry value by causing the processor to: aggregate the common format power telemetry value into an aggregated power telemetry value; and generate a power limiting management response to cause power consumption to be throttled in an integrated circuit (IC) chip based on the aggregated power telemetry value.
26. The non-transitory computer-readable medium of clause 25, wherein the computer-executable instructions further cause the processor to convert the aggregated power telemetry value to a converted power telemetry value, wherein a second unit value represented by a least significant bit of the converted power telemetry value is one (1).
27. The non-transitory computer-readable medium of clause 26, wherein the computer-executable instructions cause the processor to convert the aggregated power telemetry value to the converted power telemetry value by causing the processor to multiply the aggregated power telemetry value by the first unit value.
28. The non-transitory computer-readable medium of any one of clauses 23-27, wherein: the computer-executable instructions further cause the processor to convert the input telemetry value into the common format telemetry value by causing the processor to perform a rightward bit shift of the input telemetry value by 10 bits; and the first unit value is 1.024 based on the quotient of $2^{\text{sup.10}}/10^{\text{sup.3}}$.
29. The non-transitory computer-readable medium of any one of clauses 23-27, wherein: the computer-executable instructions further cause the processor to convert the input telemetry value into the common format telemetry value by causing the processor to perform a rightward bit shift of the input telemetry value by 20 bits; and the first unit value is 1.048 based on the quotient of $2^{\text{sup.20}}/10^{\text{sup.6}}$.
30. The non-transitory computer-readable medium of any one of clauses 23-27, wherein: the

computer-executable instructions further cause the processor to convert the input telemetry value into the common format telemetry value by causing the processor to perform a rightward bit shift of the input telemetry value by 30 bits; and the first unit value is 1.073 based on the quotient of $2^{30}/10^{9.9}$.

31. The non-transitory computer-readable medium of any one of clauses 23-30, wherein: the computer-executable instructions further cause the processor to determine whether a conversion of the input telemetry value into the common format telemetry value is required; and the computer-executable instructions cause the processor to convert the input telemetry value into the common format telemetry value responsive to determining that the conversion of the input telemetry value into the common format telemetry value is required.

Claims

1. An integrated circuit (IC) chip comprising a processor-based system, the processor-based system configured to: receive, from an input source circuit of the processor-based system, an input telemetry value; convert the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10; and process the common format telemetry value, wherein: the input telemetry value comprises an input power telemetry value; and the common format telemetry value comprises a common format power telemetry value; the input power telemetry value and the common format power telemetry value each represent a measurement of one of capacitance, electric charge, energy, and current, and wherein the processor-based system is further configured to process the common format telemetry value by being configured to: aggregate the common format power telemetry value into an aggregated power telemetry value; and generate a power limiting management response to cause power consumption to be throttled in the IC chip based on the aggregated power telemetry value.

2. The IC chip of claim 1, wherein the processor-based system is further configured to: receive the aggregated power telemetry value; and convert the aggregated power telemetry value to a converted power telemetry value, wherein a second unit value represented by a least significant bit of the converted power telemetry value is one (1).

3. The IC chip of claim 2, wherein the IC chip is configured to convert the aggregated power telemetry value to the converted power telemetry value by being configured to multiply the aggregated power telemetry value by the first unit value.

4. The IC chip of claim 1, wherein: the IC chip is configured to convert the input telemetry value into the common format telemetry value by being configured to perform a rightward bit shift of the input telemetry value by 10 bits; and the first unit value is 1.024 based on the quotient of $2^{10}/10^{3.3}$.

5. The IC chip of claim 1, wherein: the IC chip is configured to convert the input telemetry value into the common format telemetry value by being configured to perform a rightward bit shift of the input telemetry value by 20 bits; and the first unit value is 1.048 based on the quotient of $2^{20}/10^{6.6}$.

6. The IC chip of claim 1, wherein: the IC chip is configured to convert the input telemetry value into the common format telemetry value by being configured to perform a rightward bit shift of the input telemetry value by 30 bits; and the first unit value is 1.073 based on the quotient of $2^{30}/10^{9.9}$.

7. The IC chip of claim 1, wherein: the IC chip is further configured to determine whether a conversion of the input telemetry value into the common format telemetry value is required; and the IC chip is configured to convert the input telemetry value into the common format telemetry value responsive to determining that the conversion of the input telemetry value into the common

format telemetry value is required.

8. The IC chip of claim 1, wherein: the IC chip comprises a power estimation and limiting (PEL) circuit; and the input source circuit comprises one of a local activity management (LAM) circuit of the IC, a regional activity management (RAM) circuit of the IC, a processing unit of the IC, and a power management IC (PMIC) of the IC.

9. The IC chip of claim 1, integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; a drone; and a multicopter.

10. An integrated circuit (IC) chip, comprising: means for receiving, from an input source circuit, an input telemetry value; means for converting the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10; means for processing the common format telemetry value, wherein: the input telemetry value comprises an input power telemetry value; and the common format telemetry value comprises a common format power telemetry value; the input power telemetry value and the common format power telemetry value each represent a measurement of one of capacitance, electric charge, energy, and current, and wherein the means for processing the common format telemetry value comprises: means for aggregating the common format power telemetry value into an aggregated power telemetry value; and means for generating a power limiting management response to cause power consumption to be throttled in the IC chip based on the aggregated power telemetry value.

11. A method for converting a telemetry value into a common data format in a processor-based system in an integrated circuit (IC) chip, the method comprising: receiving, from an input source circuit of the processor-based system, an input telemetry value; converting the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10; processing the common format telemetry value, wherein the input telemetry value comprises an input power telemetry value; the common format telemetry value comprises a common format power telemetry value; and the input power telemetry value and the common format power telemetry value each represent a measurement of one of capacitance, electric charge, energy, and current, wherein processing the common format telemetry value comprises: aggregating the common format power telemetry value into an aggregated power telemetry value; and generating a power limiting management response to cause power consumption to be throttled in the IC chip based on the aggregated power telemetry value.

12. The method of claim 11, further comprising: receiving the aggregated power telemetry value; and converting the aggregated power telemetry value to a converted power telemetry value, wherein a second unit value represented by a least significant bit of the converted power telemetry value is one (1).

13. The method of claim 12, wherein converting the aggregated power telemetry value to the converted power telemetry value comprises multiplying the aggregated power telemetry value by the first unit value.

14. The method of claim 11, wherein: converting the input telemetry value into the common format telemetry value comprises performing a rightward bit shift of the input telemetry value by 10 bits;

and the first unit value is 1.024 based on the quotient of 2.sup.10/10.sup.3.

15. The method of claim 11, wherein: converting the input telemetry value into the common format telemetry value comprises performing a rightward bit shift of the input telemetry value by 20 bits; and the first unit value is 1.048 based on the quotient of 2.sup.20/10.sup.6.

16. The method of claim 11, wherein: converting the input telemetry value into the common format telemetry value comprises performing a rightward bit shift of the input telemetry value by 30 bits; and the first unit value is 1.073 based on the quotient of 2.sup.30/10.sup.9.

17. The method of claim 11, further comprising determining that a conversion of the input telemetry value into the common format telemetry value is required; wherein converting the input telemetry value into the common format telemetry value is responsive to determining that the conversion of the input telemetry value into the common format telemetry value is required.

18. The method of claim 11, wherein: the IC chip comprises a power estimation and limiting (PEL) circuit; and the input source circuit comprises one of a local activity management (LAM) circuit of the IC, a regional activity management (RAM) circuit of the IC, a processing unit of the IC, and a power management IC (PMIC) of the IC.

19. A non-transitory computer-readable medium, having stored thereon computer-executable instructions that, when executed, cause a processor of a processor-based device to: receive, from an input source circuit of the processor-based system, an input telemetry value; convert the input telemetry value into a common format telemetry value, wherein a first unit value represented by a least significant bit of the common format telemetry value is greater than one (1) and is based on a quotient of a power of two (2) and a corresponding power of 10; process the common format telemetry value, wherein the input telemetry value comprises an input power telemetry value; and the common format telemetry value comprises a common format power telemetry value; the input power telemetry value and the common format power telemetry value each represent a measurement of one of capacitance, electric charge, energy, and current, wherein the computer-executable instructions cause the processor to process the common format telemetry value by causing the processor to: aggregate the common format power telemetry value into an aggregated power telemetry value; and generate a power limiting management response to cause power consumption to be throttled in an integrated circuit (IC) chip based on the aggregated power telemetry value.

20. The non-transitory computer-readable medium of claim 19, wherein the computer-executable instructions further cause the processor to convert the aggregated power telemetry value to a converted power telemetry value, wherein a second unit value represented by a least significant bit of the converted power telemetry value is one (1).

21. The non-transitory computer-readable medium of claim 20, wherein the computer-executable instructions cause the processor to convert the aggregated power telemetry value to the converted power telemetry value by causing the processor to multiply the aggregated power telemetry value by the first unit value.

22. The non-transitory computer-readable medium of claim 19, wherein: the computer-executable instructions further cause the processor to convert the input telemetry value into the common format telemetry value by causing the processor to perform a rightward bit shift of the input telemetry value by 10 bits; and the first unit value is 1.024 based on the quotient of 2.sup.10/10.sup.3.

23. The non-transitory computer-readable medium of claim 19, wherein: the computer-executable instructions further cause the processor to convert the input telemetry value into the common format telemetry value by causing the processor to perform a rightward bit shift of the input telemetry value by 20 bits; and the first unit value is 1.048 based on the quotient of 2.sup.20/10.sup.6.

24. The non-transitory computer-readable medium of claim 19, wherein: the computer-executable instructions further cause the processor to convert the input telemetry value into the common

format telemetry value by causing the processor to perform a rightward bit shift of the input telemetry value by 30 bits; and the first unit value is 1.073 based on the quotient of $2^{30}/10^9$.

25. The non-transitory computer-readable medium of claim 19, wherein: the computer-executable instructions further cause the processor to determine whether a conversion of the input telemetry value into the common format telemetry value is required; and the computer-executable instructions cause the processor to convert the input telemetry value into the common format telemetry value responsive to determining that the conversion of the input telemetry value into the common format telemetry value is required.
